



SHZV Vacuum On-Load Tap-changer

Operation Instructions

HM 0.460.3901-2023

IMPORTANT:
*Please read this instruction carefully
before using the on-load tap-
changer and maintain it properly.*



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The original document is written in Chinese.

Keep this instruction at hand for future use and contact Huaming if you have any questions.

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1. Foreword

Manufacturer

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Subject to change without notice

The information contained in this technical file consists of the technical specifications published at the time printing, important modifications will be included in the new version of the technical file.

1.1 Safety

Safety instructions are explained in this instruction. They must be observed and followed to avoid relevant dangers. Personal safety equipment must be worn during work to prevent potential risks.

	Protective clothing for protecting personnel from hazards in the work environment.
	Safety shoes for protecting personnel from falling sharp objects and slippery surfaces.
	Goggles for protecting eyes and facial area from impact, dust, liquid splash and high heat hazards.
	Hard helmet for protecting head from injury.
	Hearing protection for protecting hearing from damage.
	Protective gloves for protecting hands from mechanical, chemical, and electrical hazards.

Table 1.1.1 Personal safety equipment list

1.2 Security

The following format is used for warnings in this technical paper:



warning slogan (Consequences of the violation)

Adverse consequences of safety violations

Safe behaviours to avoid adverse consequences

The following signal words are used:

Warning slogan	Hazard level	Consequences of Violations
Danger	Immediate threat of danger	Death or severe injury
Warning	Probable threat of danger	Fatal or severe injury
Prohibition	Probable dangerous situation	Minor or moderate injury
Note	Possible dangerous situation	Material damage

Table 1.2.1 Signal words in safety instructions

Pictograms warn of dangers:

Pictogram	Meaning
	The yellow triangle warning sign means life and health may be at risk if neglecting the instruction. Neglecting this warning can lead to severe or fatal injuries.
	The yellow triangle warning sign means equipment and personnel may be at risk of electricity shock if neglecting the instruction. Neglecting this warning can lead to severe or fatal injuries.
	The yellow warning triangle sign means that if the indication is ignored, equipment and personnel can be exposed to fire, deflagration, etc.
	General prohibition signs! The red symbol indicates that the behaviour prohibited.

Table 1.2.2 Symbols used in warning notices

1.2.1 Information system

Information is designed to simplify and improve understanding of particular procedures. In this technical file they are laid out as follows:



NOTE!

The blue note symbol indicates important information that must be noted.

1.2.2 Safety precautions

Precaution	Action
Electric shock	Always unplug a device before inspection. Stay away from high-voltage equipment and wear protective clothing.
Transformer oil	Collect used transformer oil in dedicated drums. Keep it away from flames or sparks. Do not touch it which may cause damages to health.
Fire	In case of fire, extinguish it with powder, foam, or carbon dioxide. Do not use water
Waste	Wastes are dangerous to the environment or human health. Clean them up in time and use different methods according to the wastes' categories

Table 1.2.3 Safety precaution

1.2.3 Appropriate operation

The product and associated equipment and special tools supplied with it comply with the relevant legislation, regulations, and standards, particularly health and safety requirements, applicable at the time of delivery.

If used as intended in compliance with the specified requirements and conditions in this technical file as well as the warning notices in this technical file and attached to the product, then the product does not present any hazards for persons, property, or the environment. This applies during the entire lifespan, from delivery through installation and operation to disassembly and disposal.

The operational quality assurance system ensures a consistently high-quality standard, particularly when it comes to observance of the health and safety requirements.

The following operations are considered as compliant operations:

- 1) The products are to be operated in accordance with this technical file, the agreed delivery conditions and the technical data, and the associated equipment and special tools supplied with the products are to be used exclusively for the intended purpose and in accordance with the specifications of this technical file
- 2) This product can only be used with the transformer specified in the order. The serial number of the motor-drive unit corresponds to that of the SHZV.

1.2.4 Inappropriate use

Operating not performed in accordance with Chapter 1.2.3 will be considered inappropriate.

Shanghai Huaming Power Equipment Co., Ltd does not accept liability for damage from unauthorized or inappropriate changes to the product. Unauthorized changes to the product without consultation with Huaming can lead to personal injury, material damage and operational faults.

1.2.5 Personnel qualification

The product is designed only for use in electrical or energy systems and facilities operated by appropriately trained staff. This staff comprises people who are familiar with the installation, assembly, commissioning, and operation of such products.

1.2.6 Obligation of the operator

To prevent accidents, faults and damage as well as unacceptable adverse effects on the environment, those responsible for transport, installation, operation, maintenance and disposal of the product or parts of the product must ensure the following:

- 1) All warning and hazard notices are complied with.
- 2) Personnel are instructed regularly in all relevant aspects of operational safety, the Operating Instructions and particularly the safety instructions contained therein.
- 3) Regulations and operating instructions for safe working as well as the relevant instructions for staff procedures in the case of accidents and fires are kept on hand at all times and displayed in the workplace where applicable.
- 4) The product is only used in a sound operational condition and safety equipment in particular is checked regularly for operational reliability.
- 5) Use only replacement parts, lubricants and aids which are authorized by the manufacturer.
- 6) Comply with the specified operating conditions and requirements of the installation location
- 7) All necessary devices and personal protective equipment for each activity are available.
- 8) The prescribed maintenance intervals and the relevant regulations are complied with.
- 9) Installation, electrical connection and commissioning of the product is only carried out by qualified and trained personnel in accordance with this technical file
- 10) The operator must ensure appropriate use of the product.

1.3 General

The SHZV on-load tap-changer is mounted on the transformer tank cover using the head flange and is connected to the motor-drive unit via its top gear, bevel gear box and driving shafts to achieve a manual, motor-driven and remote-controlled tap-change operations (see Figure 1.3.1).

The on-load tap-changer type SHZV is a combined on-load tap-changer, which consists of two main parts: the diverter switch and the tap selector (see Figure 1.3.2).

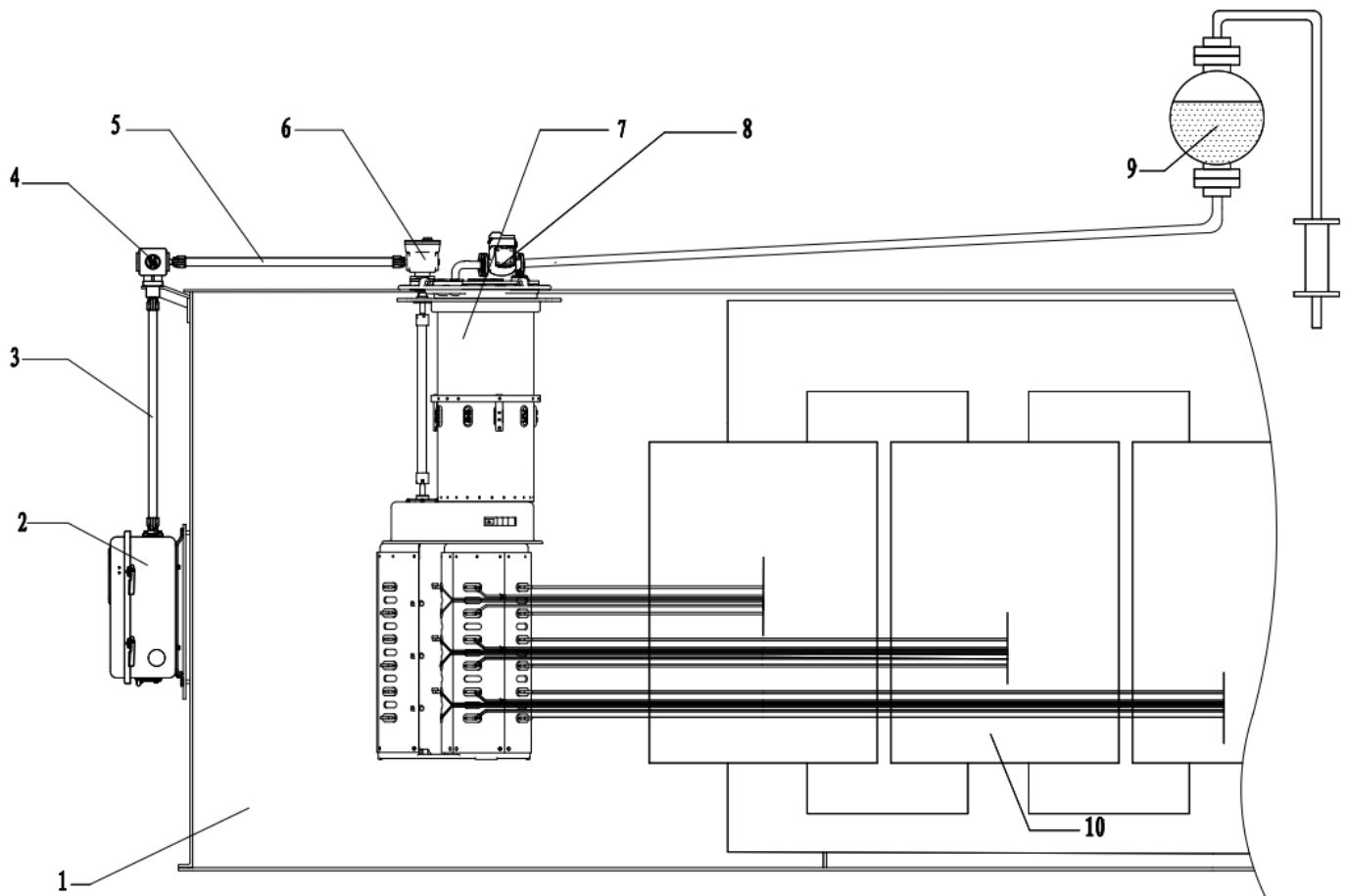


Figure 1.3.1 Overview of on-load tap-changer inside transformer

1. Transformer tank	2. Motor-drive unit
3. Vertical driving shaft	4. Bevel gear box
5. Horizontal driving shaft	6. Top gear unit
7. on-load tap-changer	8. protective relay
9. Oil conservator	10. Active part of the transformer

Table 1.3.1 List of all parts

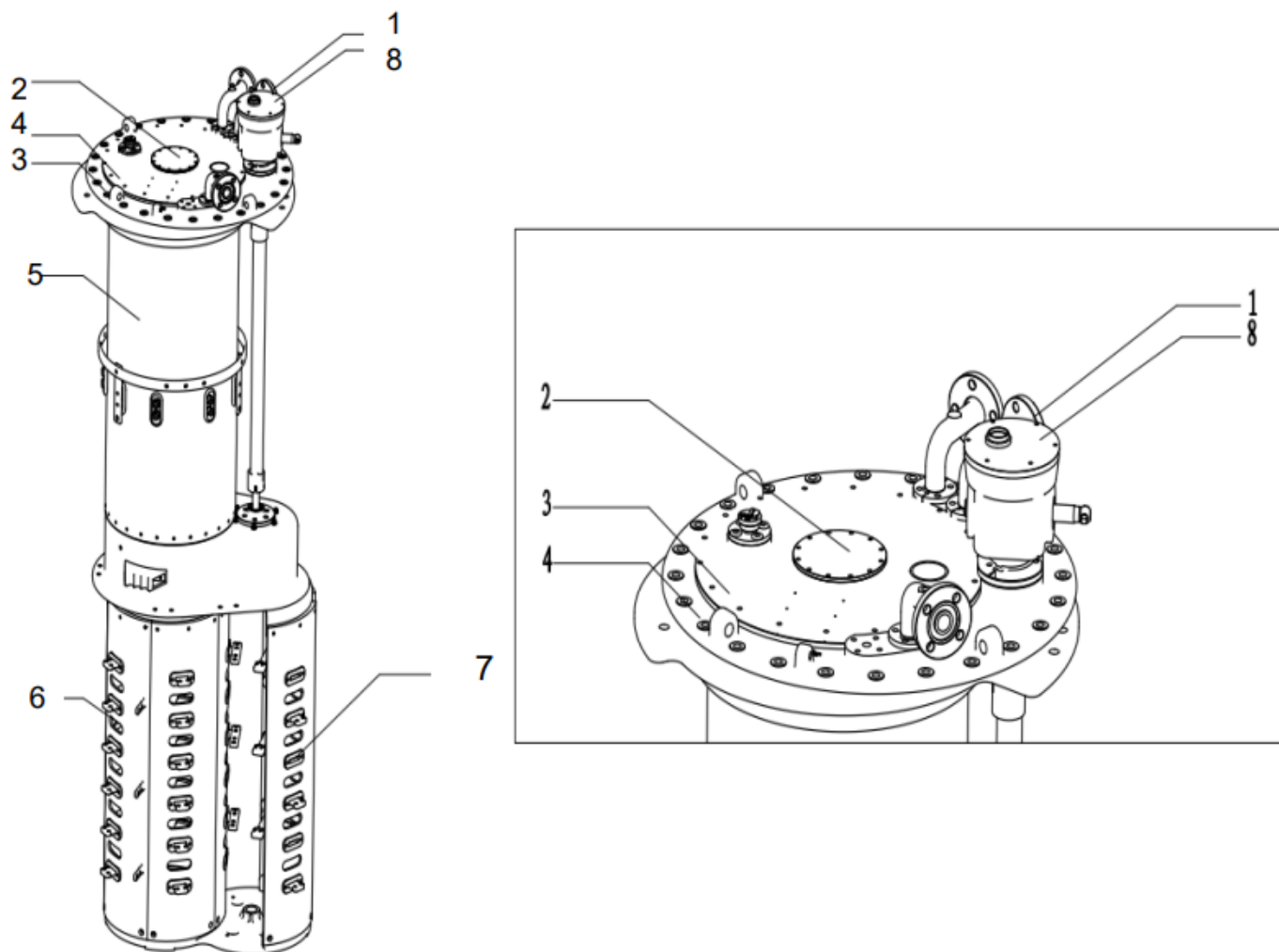


Figure 1.3.2 Overview of on-load tap-changer

1. Pipe connection	2. Rupture disk
3. on-load tap-changer head cover	4. on-load tap-changer head
5. Oil compartment	6. Change-over selector
7. Tap selector	8. Top gear unit

Table 1.3.2 List of on-load tap-changer components

2. Technical data

2.1 Technical Data

Table 2.1 Technical data

Item	Description		SHZV III			SHZV I				
1	Maximum Rated though Current (A)		400	600	1000	400	600	1000	1600	2400
2	Rated frequency (Hz)		50 or 60							
3	Connection		3-phase Y-connection for neutral point only			Single-phase for any selectable winding connection				
4	Max. rated step voltage (V)		4000							
5	Rated step capacity (kVA)		1500	1600	3000	1500	1600	3000	4400	5600
6	Short-circuit capacity (kA)	Thermal (3s)	6	8	12	6	8	12	20	24
		Dynamic (peak)	15	20	30	15	20	30	50	60
7	Max. operating positions		18 without change-over selector; 35 with change-over selector;							
8	Insulation level to earth (kV)	The highest voltage for equipment (kV)	72.5	126	170	252	300	363		
		Rated separate source AC withstand voltage (kV/50Hz, 1min)	140	230	350	460	460	510		
		Rated lightning impulse withstand voltage (kV,1.2/50μs)	350	550	750	1050	1100	1175		
9	Tap selector		Categorized into B, C, D, DE four sizes							
10	Mechanical life		1.5 million times							
11	Maintenance-free times		300,000 times							
12	Diverter switch oil compartment	Service pressure	0.03MPa							
		Leakage test	No leakage under 0.1 MPa 24 hours							
		Over pressure protection	Rupture disc bursts at 300 ± 20% MPa							
		Protective relay	Set oil flow speed at 1.0m/s ± 10							

2.2 Operation conditions

- 1) The operating temperature of the on-load tap-changer in oil is $\geq -25^{\circ}\text{C}$ and $\leq 105^{\circ}\text{C}$.
- 2) The storage environment temperature of the on-load tap-changer is $\geq -25^{\circ}\text{C}$ and $\leq 40^{\circ}\text{C}$, and the humidity should be not higher than 85%. If beyond this standard, please mark your requirements before ordering. The on-load tap-changer material and accessories will be specially processed to meet the special requirements of the environment.
- 3) After being mounted to the transformer, the non-perpendicular degree between the on-load tap-changer and the ground should be less than 2%.
- 4) No severe dust and other explosive or corrosive gases on the installation site.



Warning

Environmental Warning! (Danger of fatal or severe injury)

Condensation can cause degraded electrical performance, damage to equipment, and can also lead to explosion.

Do not store the on-load tap-changer or motor-drive unit in a place with high humidity ($\geq 85\%$). This will cause condensation.



NOTE!

Special design is available to meet particular environment conditions. Please contact Huaming for further information.

2.3 Type designation

2.3.1 Type explanation

SHZV tap-changers have a variety of specifications due to different combinations of the number of phases, the maximum rated through current, the highest voltage of the equipment, the tap selector size and the connection mode. Therefore, the above-mentioned performance parameters should be clearly reflected in the model of the on-load tap-changer, and the detailed description of each code is shown in Figure 2.3.1.

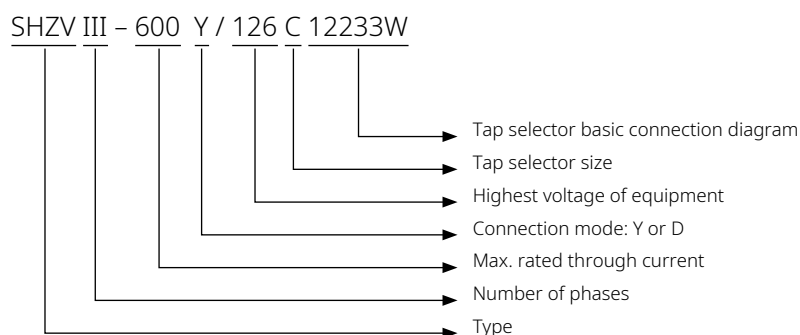


Figure 2.3.1 Type explanation

2.3.2 Basic connection

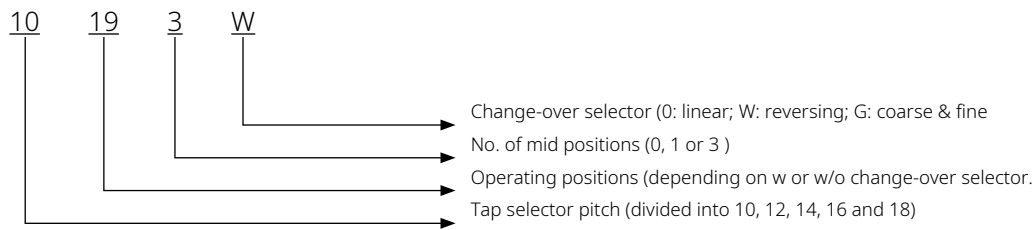


Figure 2.3.2 Tap selector basic connection

- 1) For example, 10090 means that the number of tap selector contact pitch is 10 and the maximum number of operating positions is 9, and 0 means linear regulation without change-over selector.
- 2) For example, 10193W means that the number of tap selector contact pitch is 10, a maximum number of operating positions of 19, and an middle position of 3. W represents reversing change-over selector.
- 3) For example, 10191G means that the number of tap selector contact pitch is 10, a maximum number of operating positions of 19, and an middle position of 3. G represents coarse & fine change-over selector.

3. Packing, transportation & storage

3.1 Pacing and supplied components

The complete set of on-load tap-changer contains below parts

- 1) Diverter switch
- 2) Tap selector
- 3) Motor-drive unit, controller and connecting cables, driving shafts, couplings, bevel gear box and other accessories
- 4) Protective relay and accessories
- 5) Special tools and accessories for on-load tap-changer assembly
- 6) Customer-required special accessories if any.

Please refer to the corresponding packing list delivered with the on-load tap-changer.



Figure 3.1.1 SHZV on-load tap-changer

3.2 Transportation

3.2.1 Transportation

3.2.1.1 Packings is suitable for all common kinds of transportations. The boxes of tap selector and motor-drive unit are allowed to be appropriately stacked during transportation, and the load-bearing capacity of the box's top surface shall not exceed 500kg/m².

3.2.1.2 Stacking during transportation shall be reasonably arranged according to the center of gravity marking on the box, and lifting shall be proceeded according to the lifting marks on the box.

3.2.1.3 Shipping pictograms (Figure 3.2.1)

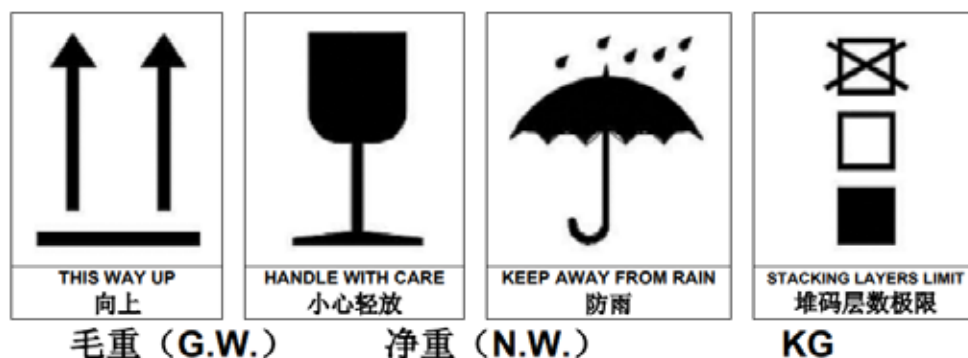


Table 3.2.1 Shipping pictograms



Danger

Danger of overturn or fall! (Danger of death or severe injury)

The selection of the sling and load reinforcement must be carried out by trained and designated personnel

Do not stand under heavy objects that are hanging.

Use transport tools and lifting devices with a bearing capacity of > 1000kg.

3.2.2 Trans-shipment

When the products arrive at the destination, the appropriate method of trans-shipment must be selected in accordance with the loading and discharge requirements (see Figure 3.2.2 and 3.2.3) :



NOTE!

Choose appropriate forklift and crane according to the dimension and gross weight of the boxes.



Figure 3.2.2 ply-wood case for forklift

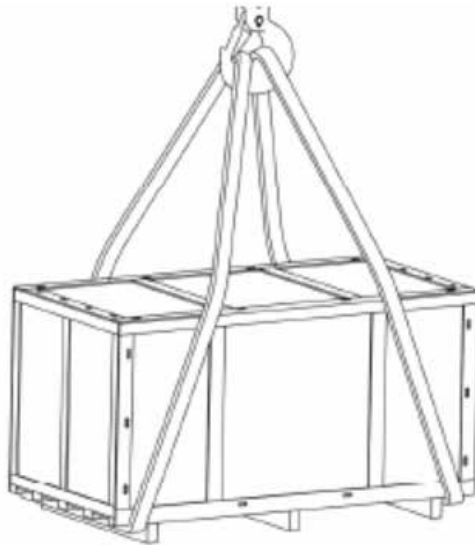


Figure 3.2.3 Ply-wood case Lifting

3.2.3 Lifting

The on-load tap-changer body should be hoisted vertically, and all lifting lugs on the top cover should be fixed before lifting. As shown below in Figure 3.2.4:

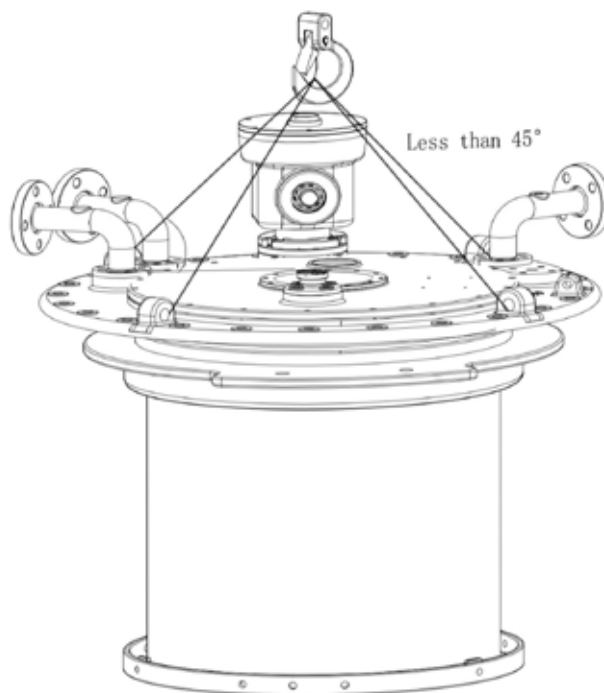


Figure 3.2.4 on-load tap-changer Lifting

Danger of lifting! (Danger of death or severe injury)

Damage to equipment or injury to personnel may happen.

At least two horizontal points must be secured during lifting or all lifting points must be fully secured before lifting.

Rapid ascending or descending is prohibited when lifting.

Lifting shall not exceed 45° to prevent overturning.

It is strictly prohibited to stand under the hoisting.



Danger

3.2.4 Goods receipt

Receipt Notes:

Each batch of delivered goods must be checked by the consignee before receipt (signing receipt note).

Following items must be checked by the consignee carefully:

- 1) Check the quantity of components received according to the packing list.
- 2) Check the technical data if it is in line with the order.
- 3) Check whether the packing is intact. If it's found to be damaged (if possible, take pictures of the box and damaged goods) or the information of the goods does not match the packing list, consignee should refuse to sign receipt note, and immediately contact with the shipping company and Huaming.



NOTE!

When receiving the goods, please check carefully the packing and the supplied components according to the packing list. In case anything is broken or missing, please contact Huaming in time.



NOTE!

The transportation and lifting of the box must be operated by professionals.
The carrying capacity of transportation vehicles and cranes should be more than 1000kg.



CAUTION!

The packing box should be firmly fixed during transportation and lifting, and be handled with care to avoid possible damage to the product due to vibration, impact, falling, tipping, bumping, shaking, etc.



CAUTION!

If the box drops or is severely impacted, the possibility of damages must be considered. Comprehensive inspection of the product should be carried out, and if necessary, inspection(repair) shall be done by the Huaming's professional and technical personnel.

The above procedure also applies to corrosion of packaged goods due to moisture (rain, snow, condensation) ingress. On-load tap-changer must be stored in a dry place and in a sealed package before installation.



NOTE!

Avoid damage of the original package when unpacking and inspecting.
If the received goods and its accessories do not match the list, contact Huaming in time.
If the box drops or is severely impacted, the possibility of damages must be considered. Comprehensive inspection of the product should be carried out, and if necessary, inspection(repair) shall be done by the Huaming's professional and technical personnel.

3.3 Storage

When the goods are stored continuously for more than 1 year, the equipment should be inspected thoroughly before installation. Specially packed goods with sealed packing can be stored outdoors if they meet the following conditions.

- 1) The storage environment temperature should not be $\geq -25^{\circ}\text{C}$ and $\leq 40^{\circ}\text{C}$. If out of this range, special material is required, please specify when ordering.
- 2) The on-load tap-changer must be stored in an environment without flammable, explosive and corrosive gases.
- 3) It must be protected from moisture, dust, rodents and ants.
- 4) Regularly check whether the stored equipment is abnormal
- 5) If the on-load tap-changer will be stored for a long time, the desiccant must be replaced regularly, and re-packed with sealed packing.

**NOTE!**

After long-term storage, the on-load tap-changer and its accessories must be carefully inspected before installation.

4. Installation of on-load tap-changer on transformer



Caution

Danger of installation! (Moderate or severe damage may occur)

Improper operation can cause damage to the on-load tap-changer

Please be sure the shielding caps to be installed on the bolts of the equipment above 35KV during installation.

4.1 Mounting flange for fixing on-load tap-changer on transformer body

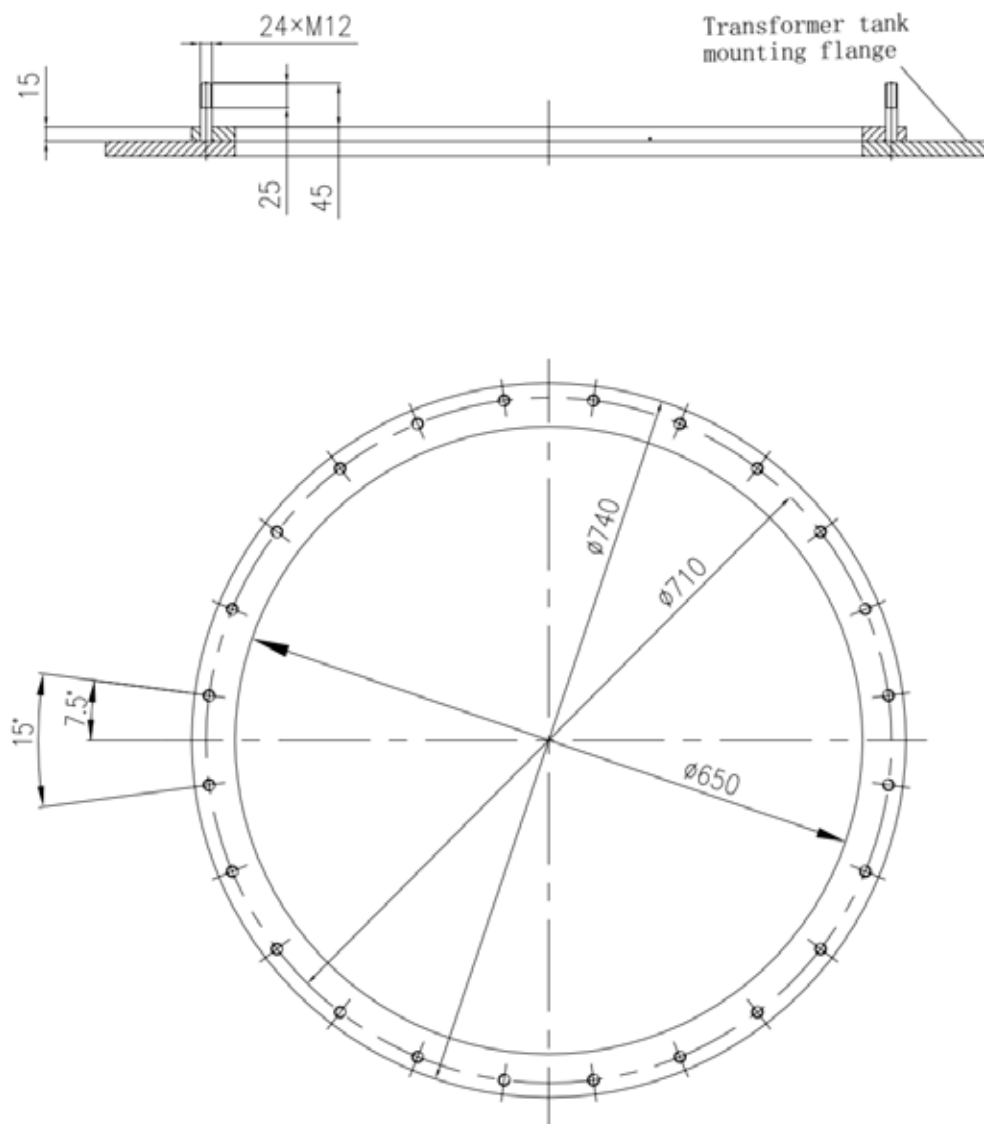


Figure 4.1.1 Mounting flanges for fixing on-load tap-changer

Mounting flanges are required for fixing the on-load tap-changer head to the transformer cover. The flange should be constructed in accordance with the shape of the sealing surface of the on-load tap-changer head (see Figure 4.1.1). The double-end screw (M12, max. length = 45 mm) should be precisely positioned.

4.2 Installation of on-load tap-changer on bell-type transformer

4.2.1 Remove the on-load tap-changer head cover and lift out diverter switch inserts.

4.2.1.1 Make sure that the on-load tap-changer is in the setting position.

4.2.1.2 Unscrew the 24 pieces of M10 bolts (with spring washers, 17mm spanner) from the head cover of the on-load tap-changer.

4.2.1.3 Remove the on-load tap-changer head cover. (see Figure 4.2.1).

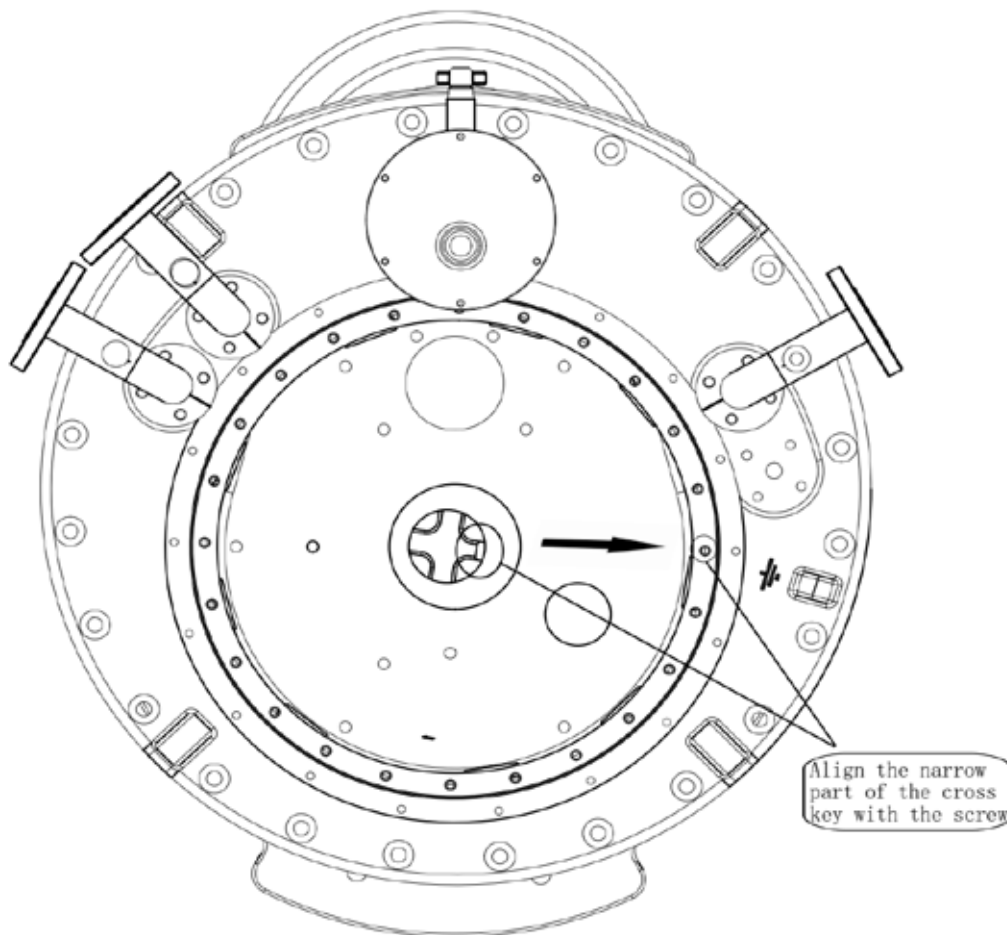


Figure 4.2.1 Schematic diagram of setting position



Danger

Danger of fall! (Moderate or severe damage may occur)

Damage to equipment and injury to personnel may result.

If the part is not placed on a flat surface, it may tilt and cause injury! Therefore they must be placed on a horizontal flat surface.

Avoid damage of the sealing surfaces of the on-load tap-changer head cover and O-ring, Figure 4.2.2) during disassembly and other operations.



NOTE!

Keep the head cover seal well preserved.

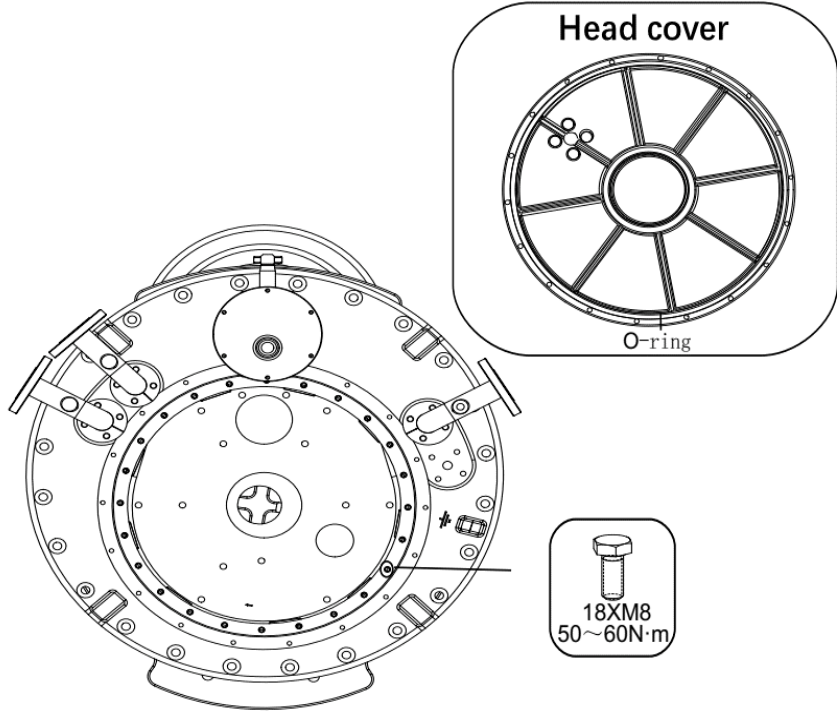


Figure 4.2.2

4.2.1.4 Remove the fastening bolts on the flange in the non-red painted area (9 pieces of M8, 13mm spanner, with spring washers) and pull out the oil suction pipe (see Figure 4.2.3).

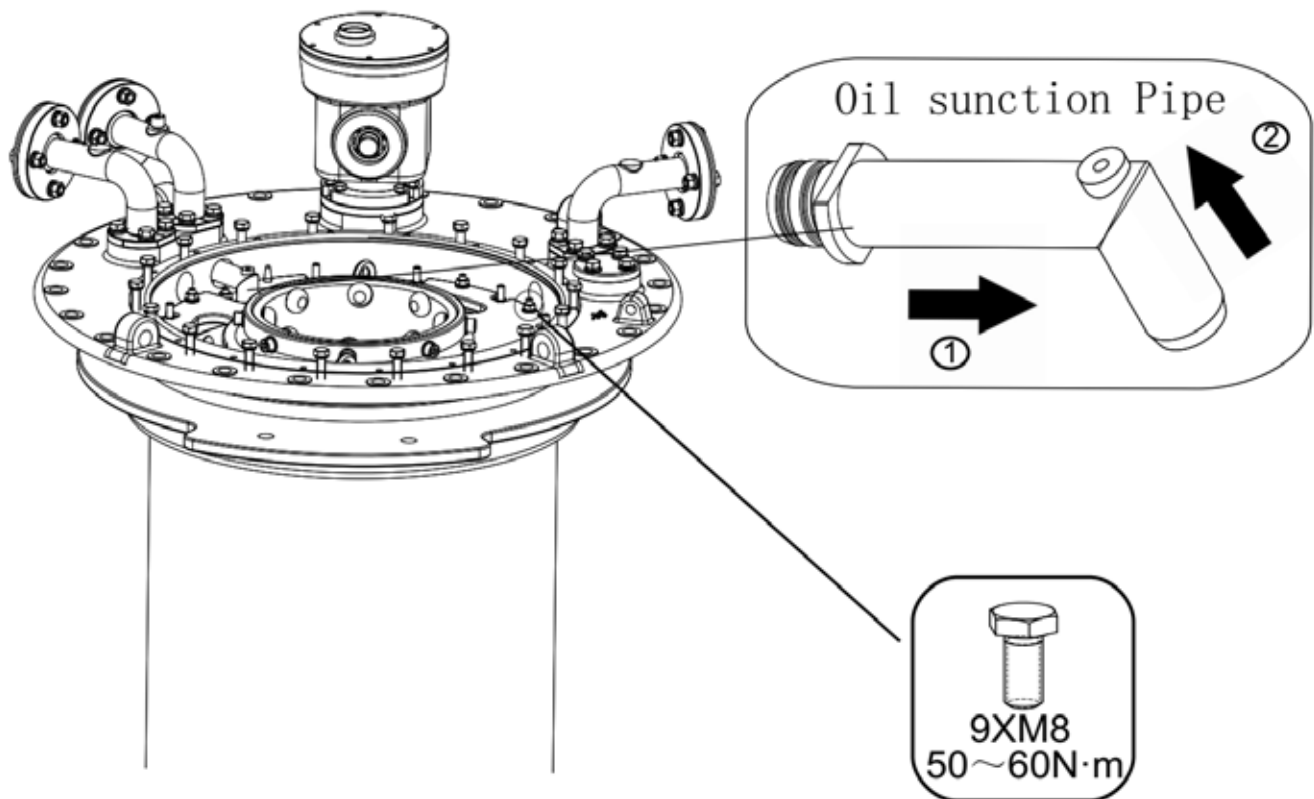


Figure 4.2.3

Caution of installation! (Damage to equipment and personnel)

Please follow the instructions below otherwise damage to the on-load tap-changer may occur. The driving shaft at the bottom of the diverter switch insert must not touch the ground and must be padded on both sides. Do not rotate the diverter switch insert before installation.



Caution

NOTE!

Do not damage the sealing gasket on the pipe connections.

Fasteners should be kept intact and not dropped into the oil compartment.

When the on-load tap-changer is tested with the transformer, the 9 pieces of M8 bolts must be fastened.



4.2.1.5 Carefully lift the diverter switch insert vertically from the oil compartment (Figure 4.2.4).

- 1) Place the diverter switch insert on a flat platform
- 2) Prevents the diverter switch insert from tilting.



After the diverter switch insert is lifted out, do not change the position of the tap selector during assembly.

4.2.2 Remove the mounting flange

Remove the remaining fixing bolts on the mounting flange (13 pieces of M8 with butterfly washers and by 13mm spanner) and lift the on-load tap-changer mounting flange from the supporting flange (see Figure 4.2.4).



NOTE!

Fasteners should be kept intact and not dropped into the oil compartment.

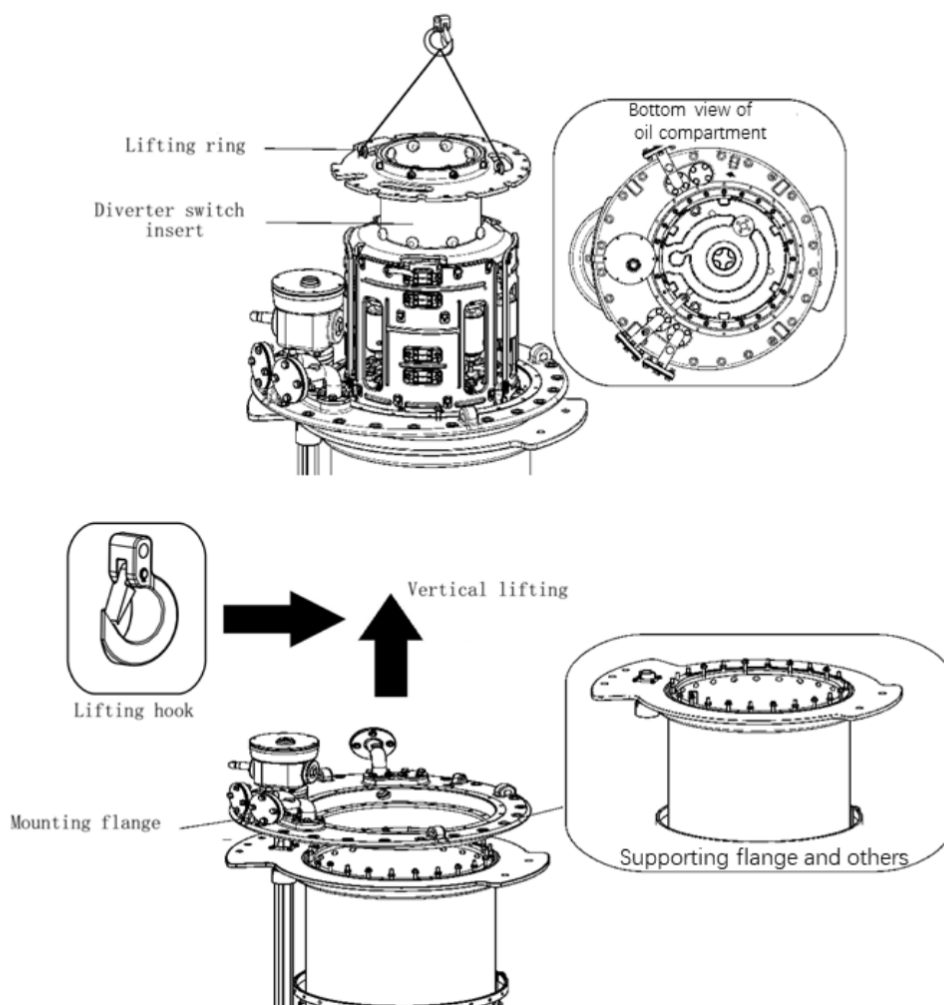


Figure 4.2.4

4.2.3 Assembly of the tap selector and diverter switch oil compartment

Lift the diverter switch oil compartment to the upper part of the tap selector and connect with the tap selector.

Proceed as follows:

4.2.3.1 Place the tap selector on a flat table.

4.2.3.2 Remove the positioning plate for tap selector. (See Figure 4.2.5)

4.2.3.3 Lift up the diverter switch oil compartment with the lifting plate. (see Figure 4.2.5).

4.2.3.4 Place the diverter switch oil compartment carefully on the upper part of the tap selector. (See Figure 4.2.5)

4.2.3.5 Align the two coupling pieces, the convex parts of upper and lower coupling pieces should be staggered when connected, be careful not to collide with each other.

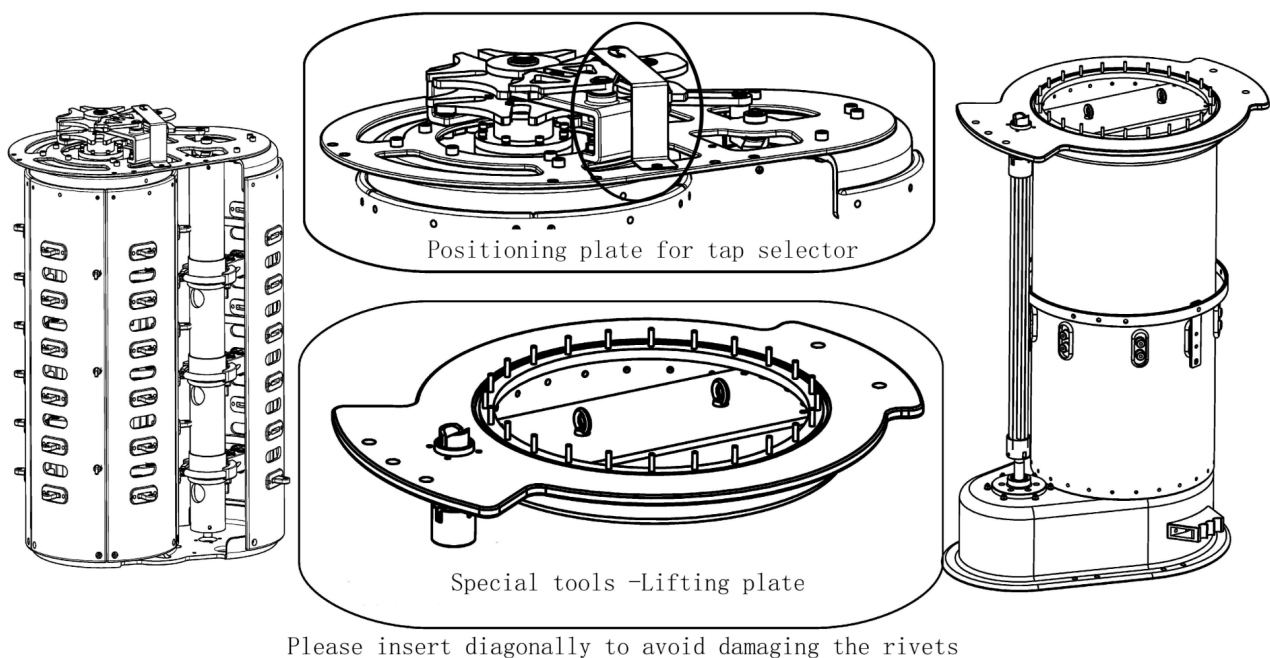


Figure 4.2.5 positioning plate, oil compartment lifting ring



Prohibition

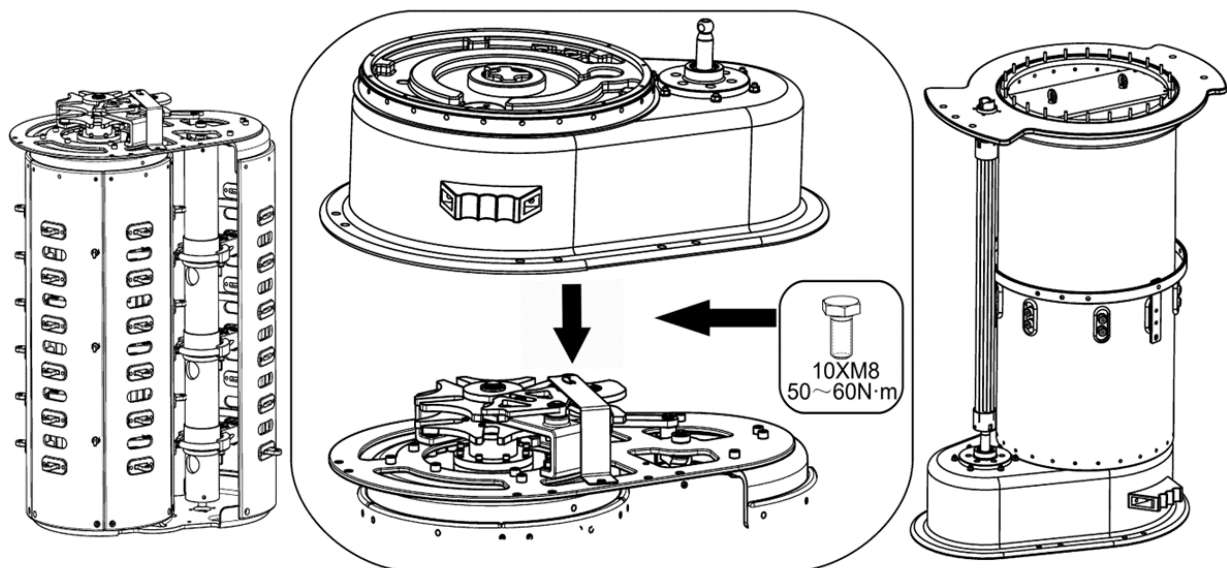
Prohibition of operation! (Damage to equipment and personnel)

Incorrect coupling will result in serious damage to the equipment!

Do not rotate the Geneva gear driving piece after removing the positioning plate.

4.2.3.6 Align the oil compartment and tap selector mounting holes, use the mounting connector in the accessory box of the tap selector to connect these two parts (see Figure 4.2.6) The torque is 50~60N·m.

4.2.3.7 Assemble the odd and even leads of tap selector and diverter switch oil compartment:
Take out the leads from the diverter switch packing box and use 25 x M8 and 35 x M10 bolts to connect the wiring terminal of diverter switch and tap selector according to the numbering.



Installation of tap selector and oil compartment

Figure 4.2.6 Tap selector and diverter switch oil compartment odd and even lead connection



Caution

Caution! (Damage to equipment and personnel)

Pay attention to the perpendicularity between the diverter switch and tap selector (error less than 2°).



Prohibition

Prohibition ! (Damage to equipment and personnel)

Do not rotate the coupling piece after removing the positioning pin.

4.2.3.8 After leads connecting, fasten the shielding cap.

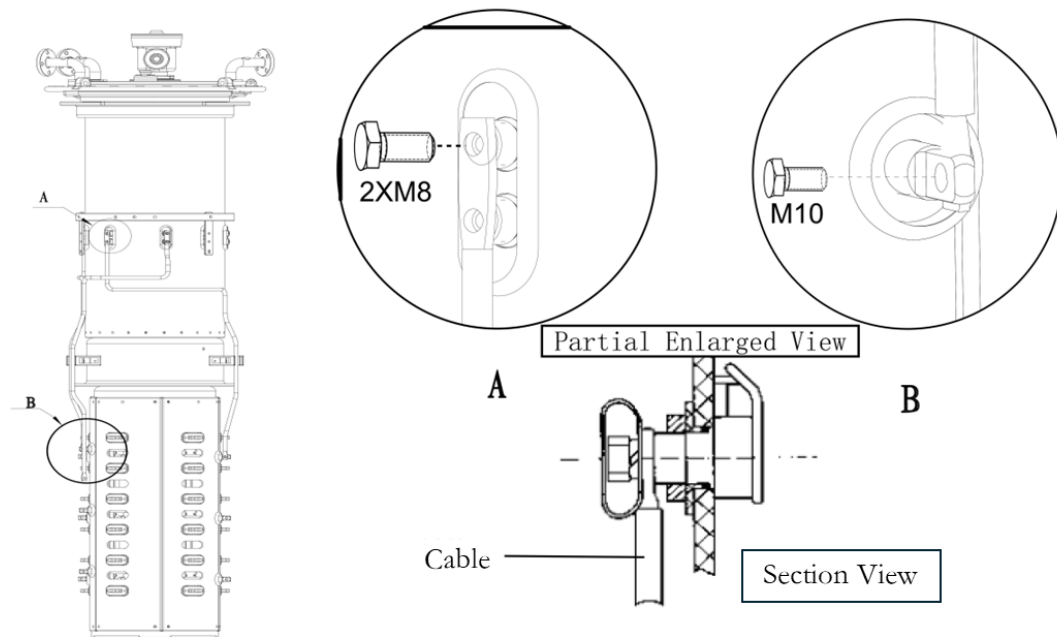


Figure 4.2.7 Tap selector and diverter switch oil compartment odd and even leads connection

Please follow the installation notes, otherwise device may be damaged.



Caution

- 1) Adhere the leads directly with the contact on the diverter switch oil compartment, and not locate the shielding cap between them (Figure 4.2.7).
- 2) The wiring of all leads must be carried out carefully. The specified torque must be used, make sure that the bolted connections are completely secure and apply the shielding caps supplied with the on-load tap-changer to terminals.
- 3) When connecting, be careful not to damage the insulation paper out of the leads.
- 4) When working on the on-load tap-changer, motor-drive unit or other components, make sure that they are securely fastened! Accidental tipping of the device must be prevented!
- 5) The on-load tap-changer must be in the accurate vertical position on the supporting structure (max. 2° deviation from vertical position). The on-load tap-changer must be mounted on the supporting structure in such a way that the on-load tap-changer can be lifted by 5 to 20 mm to its final position after the bell-type tank cover is assembled.

4.2.4 Installation of on-load tap-changer in bell-type transformer

First place the on-load tap-changer on the temporary supporting frame inside the transformer.

The assembly steps are as follows:

4.2.4.1 Lift the on-load tap-changer onto the supporting frame and connect it to the voltage regulating winding.

The on-load tap-changer shall be placed stably on the support frame and must not be moved around.

4.2.4.2 Temporarily fix the on-load tap-changer to the supporting frame using the mounting holes on the supporting flange. If necessary, put temporary adjusting spacers between the temporary supporting frame and the supporting flange and adjust the mounting height of the on-load tap-changer to the final mounting position. The adjusting spacers must be removed before installation of the transformer bell-type tank cover.

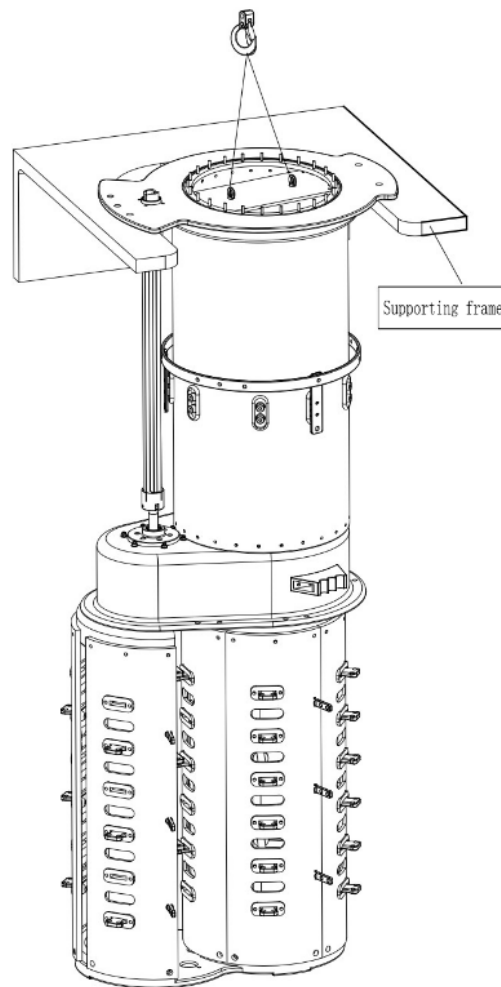


Figure 4.2.8

4.2.5 Assembly of bell-type transformer tank cover

4.2.5.1 Before assembling the bell-type transformer tank cover, clean the sealing surfaces of the supporting flange of the on-load tap-changer oil compartment. Place the sealing ring on the supporting flange (Figure 4.2.9). Pull out the adjusting spacers.

4.2.5.2 Lift the bell-type transformer tank cover over the transformer body and assemble it.

4.2.5.3 Clean the sealing surface before mounting the on-load tap-changer mounting flange. Place the sealing gasket on the transformer mounting flange and fix the on-load tap-changer mounting flange to the transformer mounting flange (24 pieces of M8 bolts, torque 100 to 110 N·m). (see Figure 4.2.10).

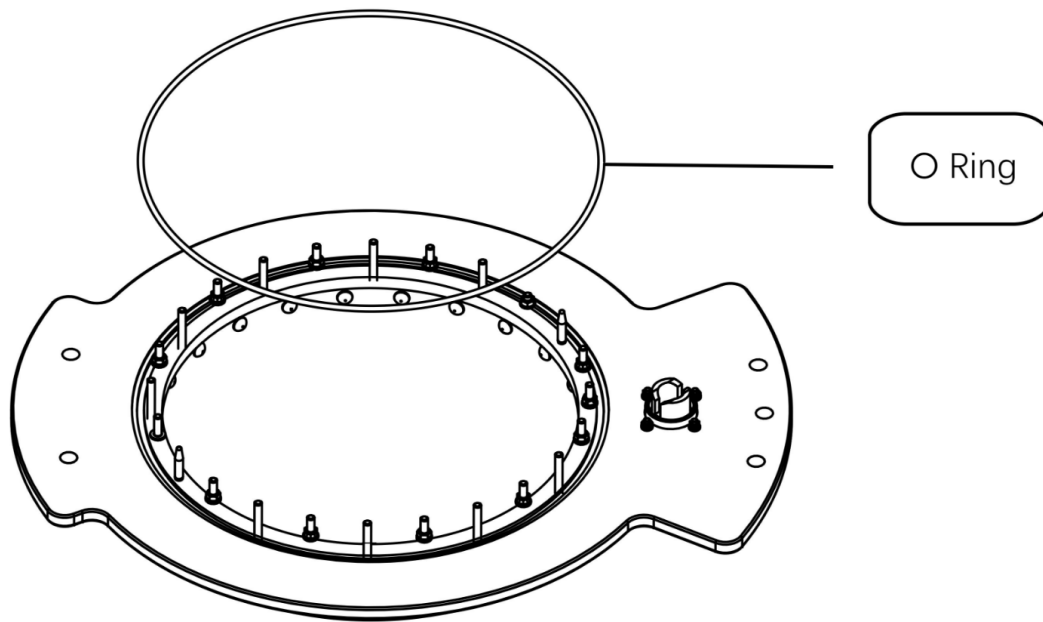


Figure 4.2.9

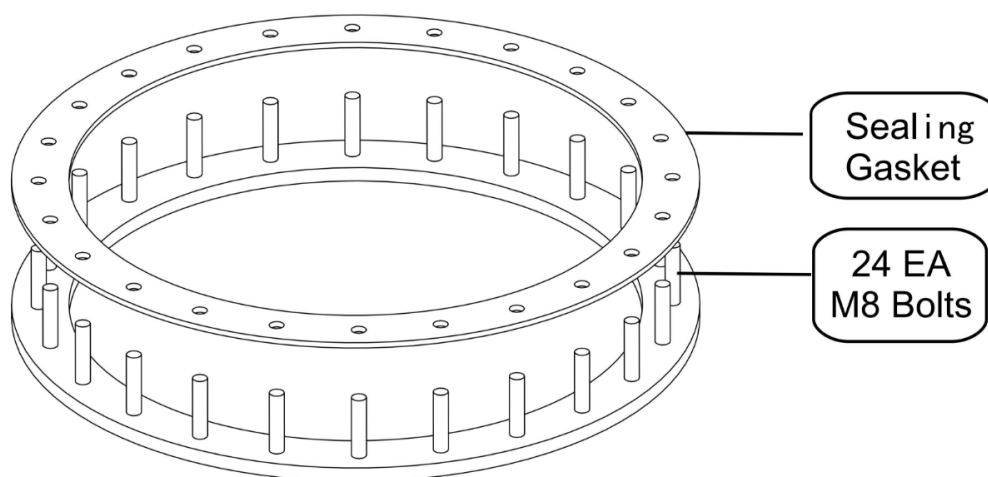


Figure 4.2.10

4.2.5.4 Lift the mounting flange onto the on-load tap-changer . Make sure that all screws of the supporting flange are accurately positioned in the fixing holes of the on-load tap-changer (see Figure 4.2.11).

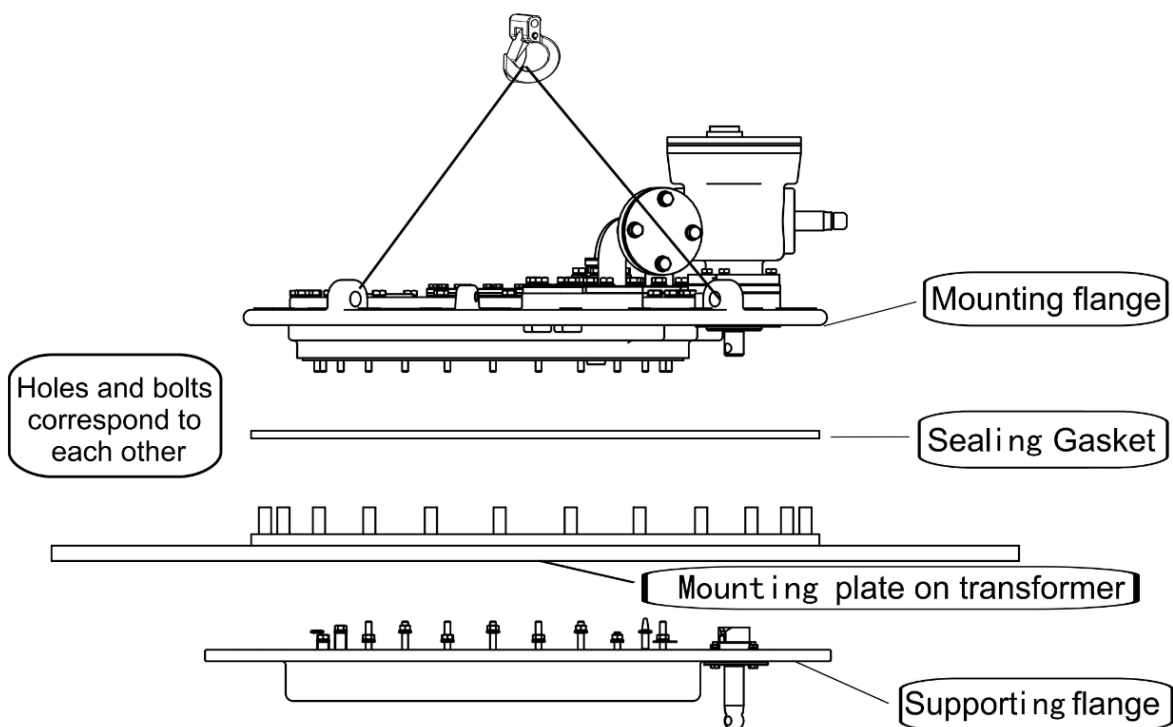


Figure 4.2.11

4.2.5.5 Mount the on-load tap-changer on the head mounting flange. Take care that the two positioning screws (extensions) are positioned and that the red triangular marks on the mounting flange and the on-load tap-changer top cover correspond with each other to ensure that the on-load tap-changer is mounted in the correct position. (see Figure 4.2.12).

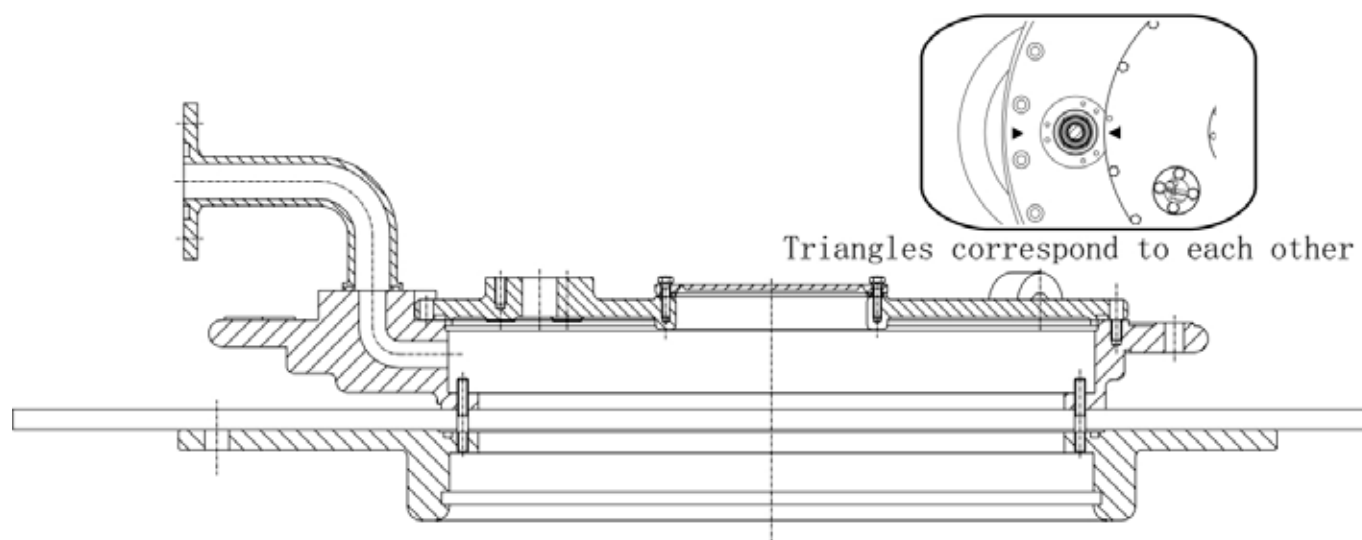


Figure 4.2.12

4.2.6 Installation of diverter switch inserts

4.2.6.1 Make sure that the tap selector and the connecting part at the bottom of the oil compartment must be in the setting position when the diverter switch insert is installed, roughly align with the position of the small end of the connecting part. (see Figure 4.2.13)

4.2.6.2 Lift the diverter switch insert above the diverter switch oil compartment.

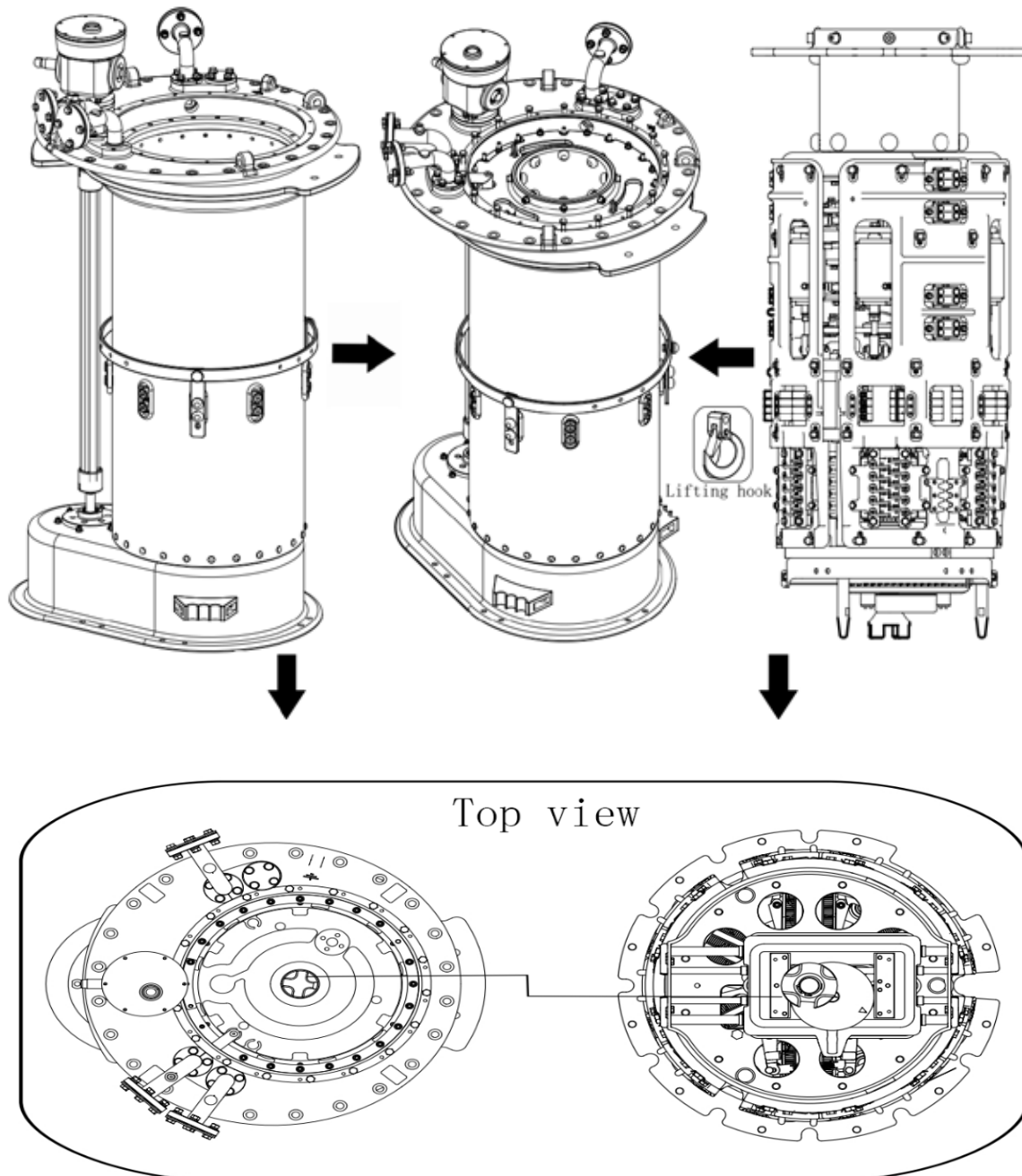


Figure 4.2.13

NOTE!



After installation of the insert, check that the lower surface of the insert flange and the positioning surface of the bell-type flange must be tightly attached without gap, otherwise it will result in serious problems such as not switching and the wrong gear!

After the insert is installed, the upper surface of the flange shall not exceed the red line.

4.2.6.3 Installation of oil suction pipe: Insert one end of the oil suction pipe into the insert's oil suction pipe positioning hole, and the other end into the bell-type flange oil suction pipe hole. (See Figure 4.2.15)

4.2.6.4 Installation of head cover: Place the head cover over the on-load tap-changer head flange to seal it. Make sure that the head cover seal in correct position.



NOTE!

Damage may be caused.

- Do not damage the O-ring seal on the pipe fitting
- Carefully clean the sealing surface before installing the head cover.

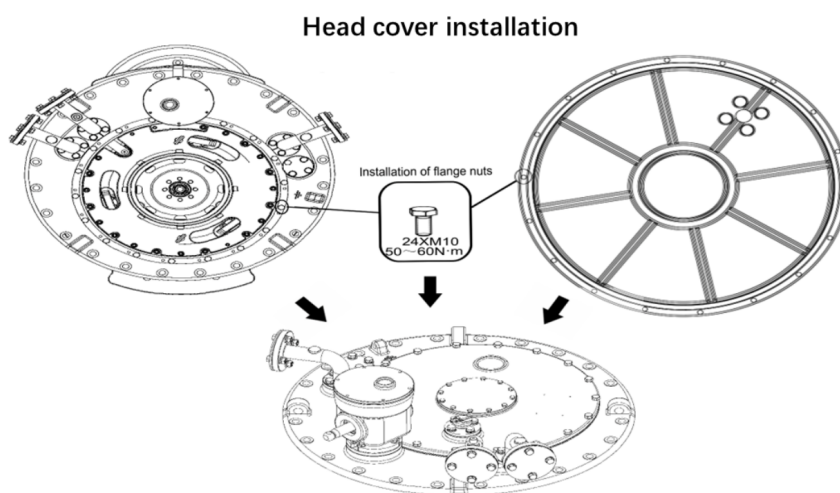


Figure 4.2.14

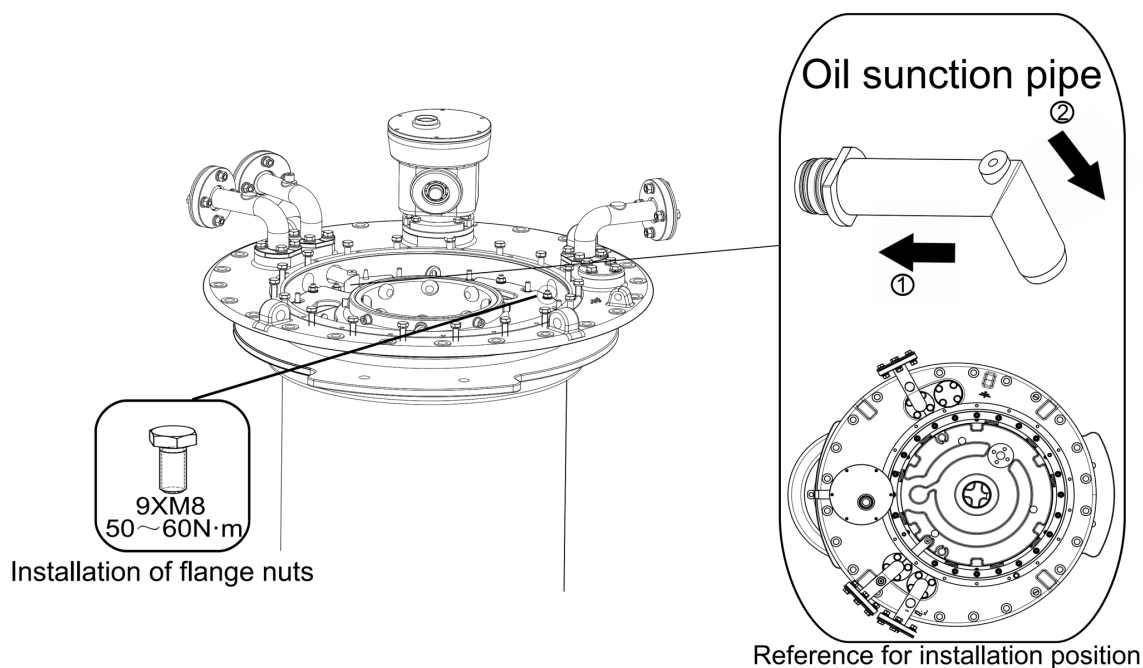


Figure 4.2.15

4.3 Installation of on-load tap-changer on standard type transformer

Please refer to chapter 4.2.1

4.4 Connection between tap leads of voltage regulating winding and on-load tap-changer terminals

The tap selector terminals are provided with through-holes for M10 bolts, which facilitate the connection of tap leads of voltage regulating winding with on-load tap-changer terminals.

Shielding caps for M10 bolts will be supplied (Figure 4.4.1).

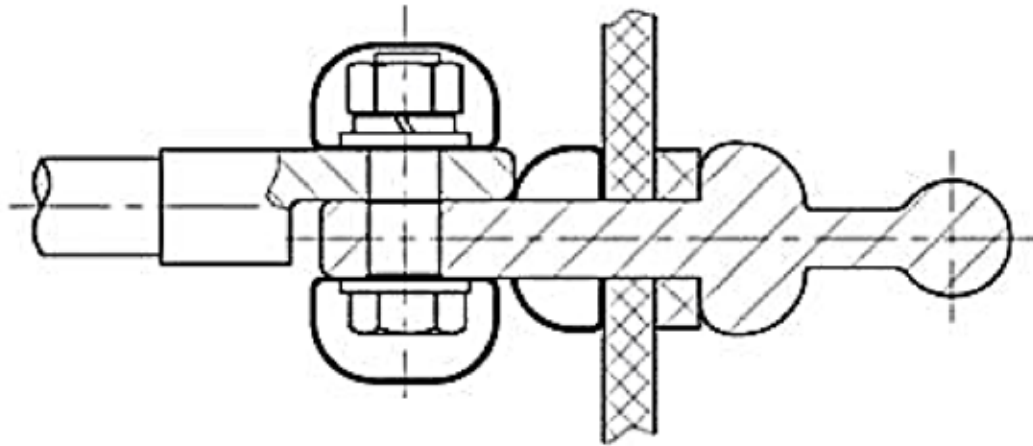


Figure 4.4.1

4.4.1 Tap leads connected to tap selector terminals shall not deform or damage the tap selector by pulling force

- 1) The tap leads of the regulating winding shall be lead to tap selector from both sides respectively, to avoid unilateral lead force, resulting in distortion of the tap selector.
- 2) The connection between the tap selector terminal and the terminal clamp of the transformer lead should have a certain deflection, which should not be too short, the connecting leads should be soft, it is recommended that the section of the connecting lead is not coated with insulating varnish, to avoid hardening and deformation of the insulating rod after drying.
- 3) The end of the leads connecting the tap selector shall be connected in the shape of expanding ring (looped), so that the tap selector insulation board is not subjected to tension.
- 4) Tap selector terminal leads should be led out from the outside of the tap selector and shall never pass through the inside of the tap selector.
- 5) The bell-type on-load tap-changer must be lifted by 5 to 20 mm after the leads are connected. For this reason, the tightness of the connecting leads should be paid special attention. It is suggested that the spacer should be located on the supporting frame, which can temporarily raised up the supporting flange to the final mounting level then connect the tap leads(same as real mounting). After the completion of the connection, remove the spacer, check the tightness of the leads and whether the on-load tap-changer is applied by pulling force.



Caution

Caution of installation! (Damage to equipment and personnel)

Improper operation can cause damage to the on-load tap-changer

Shielding caps supplied must be installed on the screws for Um 35KV and above.

Must use the bolts with proper length.

5. Transformer ratio test and DC resistance measurement

It is recommended to carry out ratio test and DC resistance measurement prior to drying treatment.

NOTE!

Operation without oil will cause damage to on-load tap-changer.

More than 10 cycles operation without oil is not allowed.

For the first operation of tap-changer after transformer being dried, make sure tap selector is fully immersed with oil and as well as the diverter switch oil compartment!

Incomplete on-load tap-changer operation will cause damage of the on-load tap-changer!



5.1 Ratio test (upper head flange and diverter switch insert are required for bell type on-load tap-changer)

NOTE!

For bell-type on-load tap-changer, mounting flange, diverter switch and top cover need to be assembled temporarily. (please refer to Figure 5.1.1)

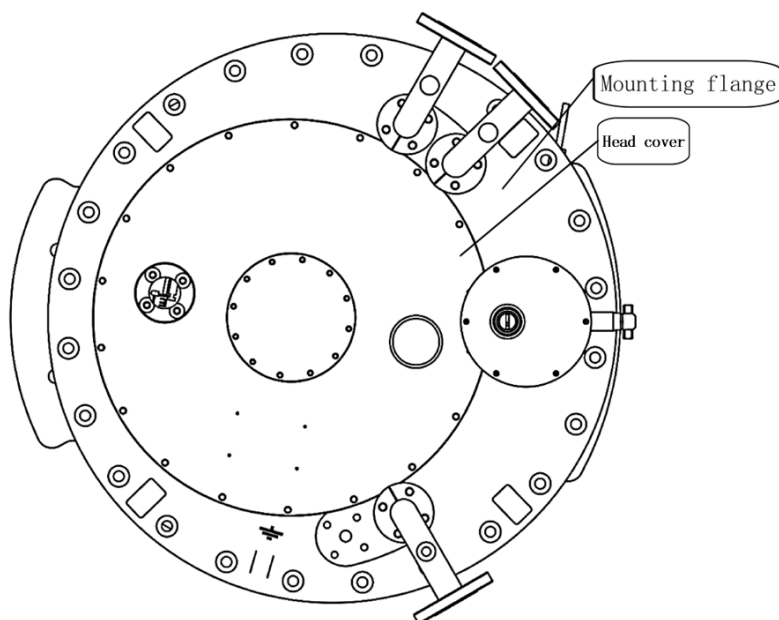


Figure 5.1.1

5.1.1 Turn the driving shaft of the on-load tap-changer top gear unit, a short tube with an inner diameter of 25mm (a diameter of 12mm) can be used. Connection bolts have been screwed into the tube), and then equipped with a hand wheel or hand crank.

5.1.2 For a group of two or three heads tap changers, all on-load tap-changer heads should be connected to each other with horizontal shafts

5.1.3 For one tap-change, the sound of the diverter switch action is clearly audible. After the sound of the diverter switch be heard, it must continue to rotate two and half turns in the same direction to complete one tap-change.

5.1.4 During operation, each tap position should be observed at any time through the observation window on the on-load tap-changer head to avoid over-riding the limit position of the on-load tap-changer.

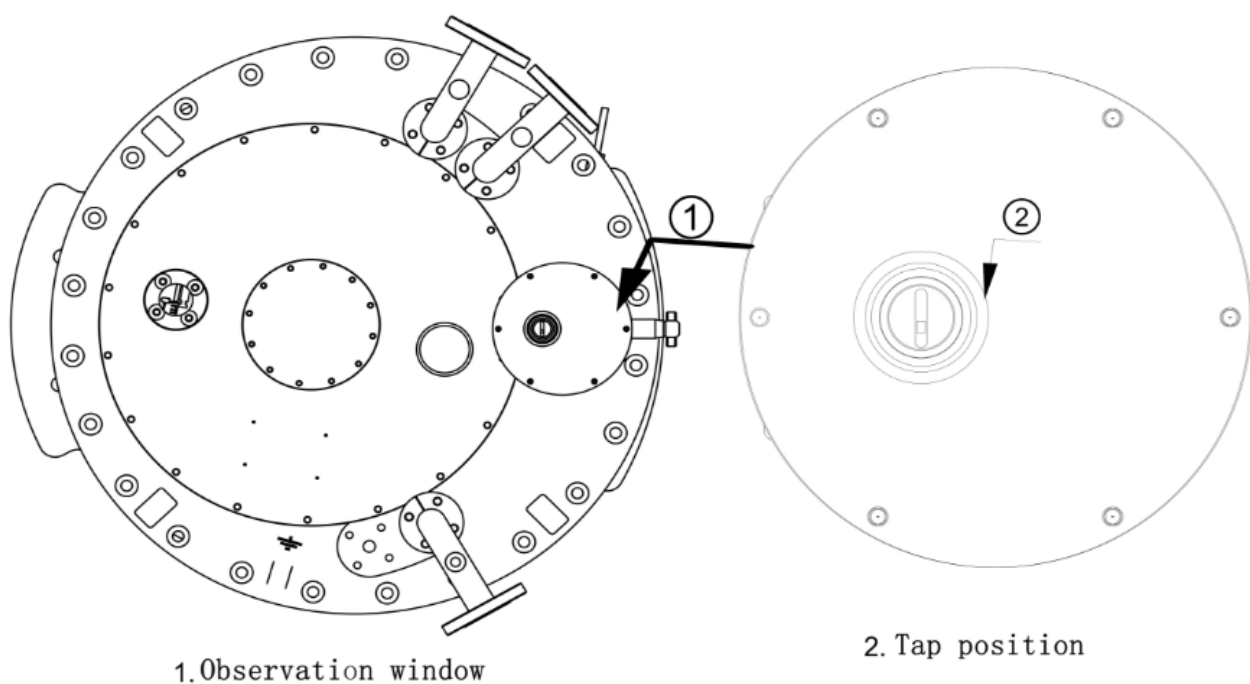


Figure 5.1.2

5.2 Transformer DC resistance measurement

Kindly note that the on-load tap-changer is operated without oil when performing transformer DC resistance measurements, minimizing the operation of the on-load tap-changer is highly recommended. After the transformer ratio test and DC resistance measurement, the on-load tap-changer must be adjusted to the setting position.

6. Drying process and oil filling

6.1 Drying process

In order to guarantee the insulation performance of on-load tap-changer, the necessary drying must be carried out according to the following instructions (vacuum drying or vapor phase drying).



Caution

Caution of drying! (Damage to equipment and personnel)

The parts, as below must be removed before the drying process: on-load tap-changer top cover, O ring, top gear unit, motor-drive unit, protective relay, pressure relief device, bevel gear box, and all accessories.

6.1.1 Vacuum drying

6.1.1.1 Drying in the oven

- a. The on-load tap-changer is heated up at a speed of 10°C/ hour in the normal pressure air to the final temperature of 110°C.
- b. Drying is carried out in the circulating hot air for 20 hours with highest temperature of 110°C.
- c. Vacuum drying lasts 50 hours, with temperature of 105°C to Max. 125°C. Maximum residual pressure 133Pa.

6.1.1.2 Drying in the transformer oil tank

- a. Connect E2 and R connections on on-load tap-changer head with by-pass connection (nominal diameter 25mm).
- b. The on-load tap-changer is heated up at a speed of 10°C/ hour in the normal pressure air, to the final temperature of 110°C.
- c. Drying is carried out in the circulating hot air for 20 hours under highest temperature of 110°C.
- d. Vacuum drying lasts 50 hours, with highest temperature of 105°C to Max.110°C. Maximum residual pressure 133Pa.

6.1.2 Vapor phase drying



CAUTION!

on-load tap-changer top cover, top gear unit and accessories are not allowed to be vacuum dried together with the transformer in the vacuum tank.

6.1.2.1 Vapor phase drying in the vacuum tank



NOTE!

Head cover, top gear unit and accessories of the on-load tap-changer are not allowed to be dried in the vacuum tank together with the transformer.

Before drying process, on-load tap-changer top cover, top gear unit, motor-drive unit, protective relay, pressure relief device, bevel gear box, temperature sensor, and other accessories must be disassembled and placed outside of vacuum tank. If it is vapor phase drying, be sure to open the oil drain bolt at the bottom of the oil compartment before drying, to discharge the condensation from the oil compartment.

6.1.2.1.1 Loosen oil drain plug

Use a special wrench to loosen the oil drain plug in the direction of counterclockwise, and the loosen of the oil drain plug is not allowed to be more than 10 turns.

The key points of loosening oil drain plug are as follows:



- 1) Unscrew oil drain plug between the oil compartment bottom and the tap selector gears clockwise. (See Figure 6.1.1)
- 2) Remove the on-load tap-changer top cover and store it outside the vacuum tank.
- 3) Remove all accessories and store them outside the vacuum tank.
- 4) Supply kerosene vapor at temperature of around 90°C and keep this temperature constant for 3 to 4 hours.
- 5) Increase the kerosene vapor temperature by 10°C/hour to the specified final temperature, but maximum temperature on the on-load tap-changer cannot exceed 125°C.
- 6) Vacuum dry tap changer at the temperature between 105°C and 125°C. The highest residual pressure is no more than 133Pa. The drying duration is the same as the transformer, but at least 50 hours.
- 7) Close the oil drain plug (torque 18~20N·m) after drying process.

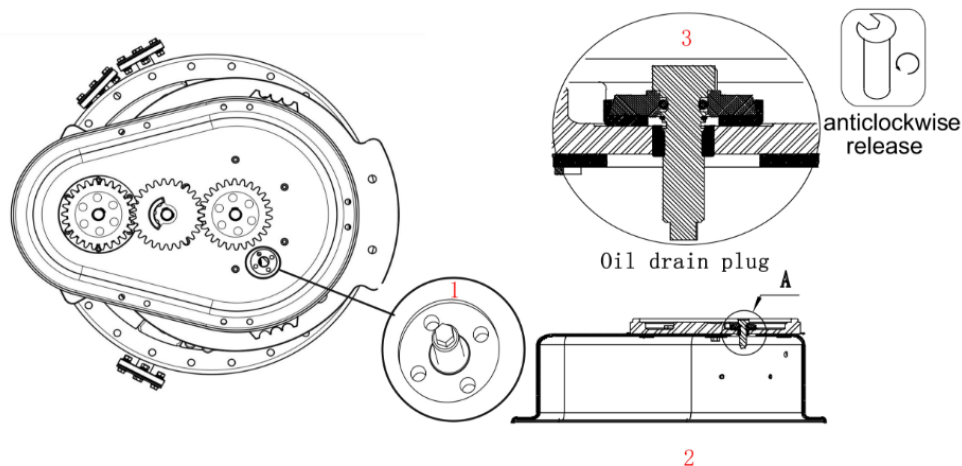


Figure 6.1.1



Caution

Caution of operation! (Damage to equipment and personnel)

Use a special wrench to loosen the oil drain plug counterclockwise, the oil drain plug should not be loosened more than 10 cycles.

6.1.2.2 Vapor phase drying in the transformer tank

Before drying in the transformer tank, the diverter switch insert should be lifted out, and the oil drain plug at the bottom of the oil compartment should be loosened with a special wrench , in order to discharge the condensation from the oil compartment.

6.1.2.2.1 Unnecessary damage to equipment may occur:



Caution

Caution of operation! Damage to equipment

- Make sure to prevent small parts from falling into the oil compartment.
- When disassembling and reassembling, the quantity of small parts must be counted clearly.
- Avoid damaging the top cover of the on-load tap-changer and the sealing surface of the on-load tap-changer head during disassembly and other operations.
- Broken seals will result in sealing-failure of oil tank which can lead to oil leakage and damage to the on-load tap-changer.
- An open air oil drain plug means that oil will escape from diverter switch oil compartment and therefore cause damage to the on-load tap-changer and transformer.

6.1.2.2.2 Main points for operation



- Avoid damaging the on-load tap-changer head cover and the sealing surfaces of on-load tap-changer head during disassembly and other operations.
- Before first operation of the on-load tap-changer after the transformer drying, make sure that the tap selector is completely immersed in transformer oil and that the diverter switch compartment is filled with oil.
- If the tap changer is drying in the transformer tank, loosen the drain plug.
- After drying, close the oil drain plug, and then refit the diverter switch insert.

6.1.2.2.3 Supply kerosene vapor to on-load tap-changer

The connections E2 and R on the on-load tap-changer head (See Figure 6.1.2) are connected by bypass connection, and the inner diameter is at least 50mm , then supply kerosene vapor.

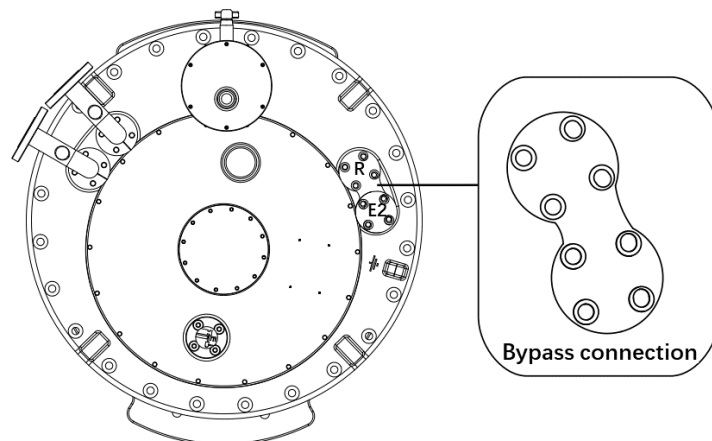


Figure 6.1.2

6.1.2.2.2 Drying

- 1) Supply kerosene vapor at a temperature of around 90°C and keep this temperature constant for 3 to 4 hours.
- 2) Increase the kerosene vapor temperature by 10°C/hour to the specified final temperature, but the maximum temperature on the on-load tap-changer cannot exceed 125°C.
- 3) In vacuum drying, the temperature on the on-load tap-changer is between 105°C and 125°C. The highest residual pressure no more than 133Pa. The drying duration is the same as the transformer, but at least 50 hours.

6.1.2.2.3 Remove the diverter switch insert and close the oil drain plug.

- 1) Remove the diverter switch insert according to chapter 4.2.1.
- 2) Screw the oil drain plug clockwise with an extended T-wrench. (See Figure 6.1.2) (Torque 20 N.m.)

6.1.2.2.4 After tightening the oil drain plug, reinstall the diverter switch insert according to chapter 4.2.6.

6.2 Oil filling

After drying, in order to prevent too much moisture from entering the oil compartment, the oil compartment (diverter switch insert already installed) should be refilled with oil as soon as possible.

The top cover of the on-load tap-changer should be closed again and 18 pieces of M8 bolts should be tightened, paying attention to the correct position of the O-ring. Both the transformer and the diverter switch are filled with oil under vacuum. In order to vacuumize the diverter switch oil compartment and the transformer simultaneously, the by-pass pipe provided by our factory can be installed between connection of R and E2 on the on-load tap-changer head.



- 1) In order to vacuumize the on-load tap-changer and the transformer at the same time, connect a by-pass connection between E2 and R connections.
- 2) Both the on-load tap-changer and the transformer should be filled with new oil under vacuum at the same time.
- 3) Only new transformer oil with standard is allowed to fill for diverter switch oil compartment and conservator.

7. Installation of on-load tap-changer protective device and more-drive unit

7.1 Installation of protective relay

Install according to the instruction of protective relay.

7.2 Installation of pressure relief device

Install according to the instruction of pressure relief device.

7.3 Installation of motor-drive unit

The motor-drive unit is used to drive the on-load tap-changer to change taps; it can be operated electrically or manually.



Attentions should be paid when installing the motor-drive unit:

7.3.1 Connect motor-drive unit with on-load tap-changer when both of them are in setting position.

7.3.2 The motor-drive unit must be mounted at the provided place on the transformer tank in a vertical position, deflection is not allowed;

7.3.3 The mounting plate for the motor-drive unit must be flat to avoid distortion and deformation of the motor-drive unit, which could affect its functionality.

7.3.4 For detailed installation instructions regarding the motor-drive unit , please refer to motor- drive unit operation instruction provided by Huaming.

7.4 Installation of bevel gear box

Bevel gear box is secured to the bracket on the transformer tank cover with two bolts (hole diameter 18mm).

NOTE!



- The serial number marked on the nameplate of bevel gear box must be consistent with the serial number on the nameplate of the on-load tap-changer.
- When horizontal and vertical driving shafts are over two meters, an intermediate gear must be applied. The installation of the specially designed bevel gear box, top gear unit and the intermediate gear of the vertical or horizontal driving shaft can be carried out according to the above instructions.

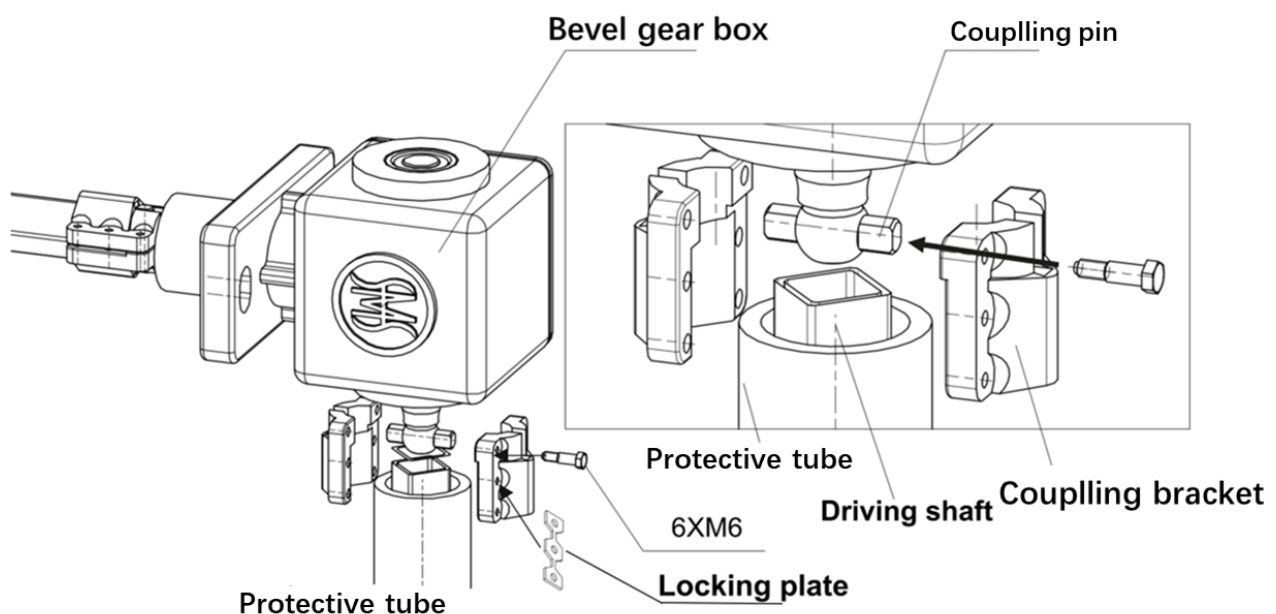


Figure 7.4.1

7.5 Installation of horizontal and vertical driving shafts

7.5.1 Connect the driving shafts, motor-drive unit, and on-load tap-changer as per the instructions specified in motor-drive unit operation instruction (Figure 7.5.1 Appendix 8)

7.5.2 Intermediate gear should be applied if the length of vertical shaft exceeds 2M. (See Figure 7.5.2)

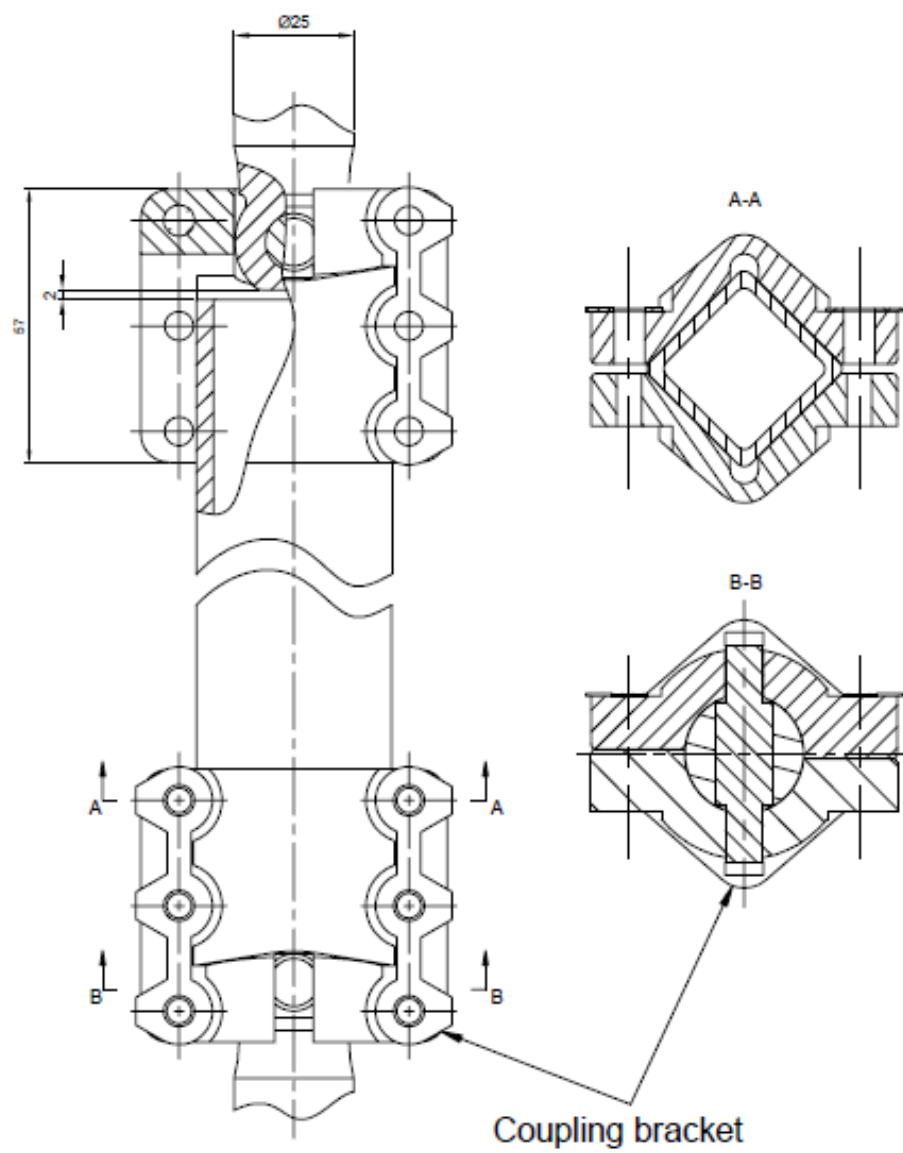


Figure 7.5.1 Driving shaft mounting dimension unit: mm

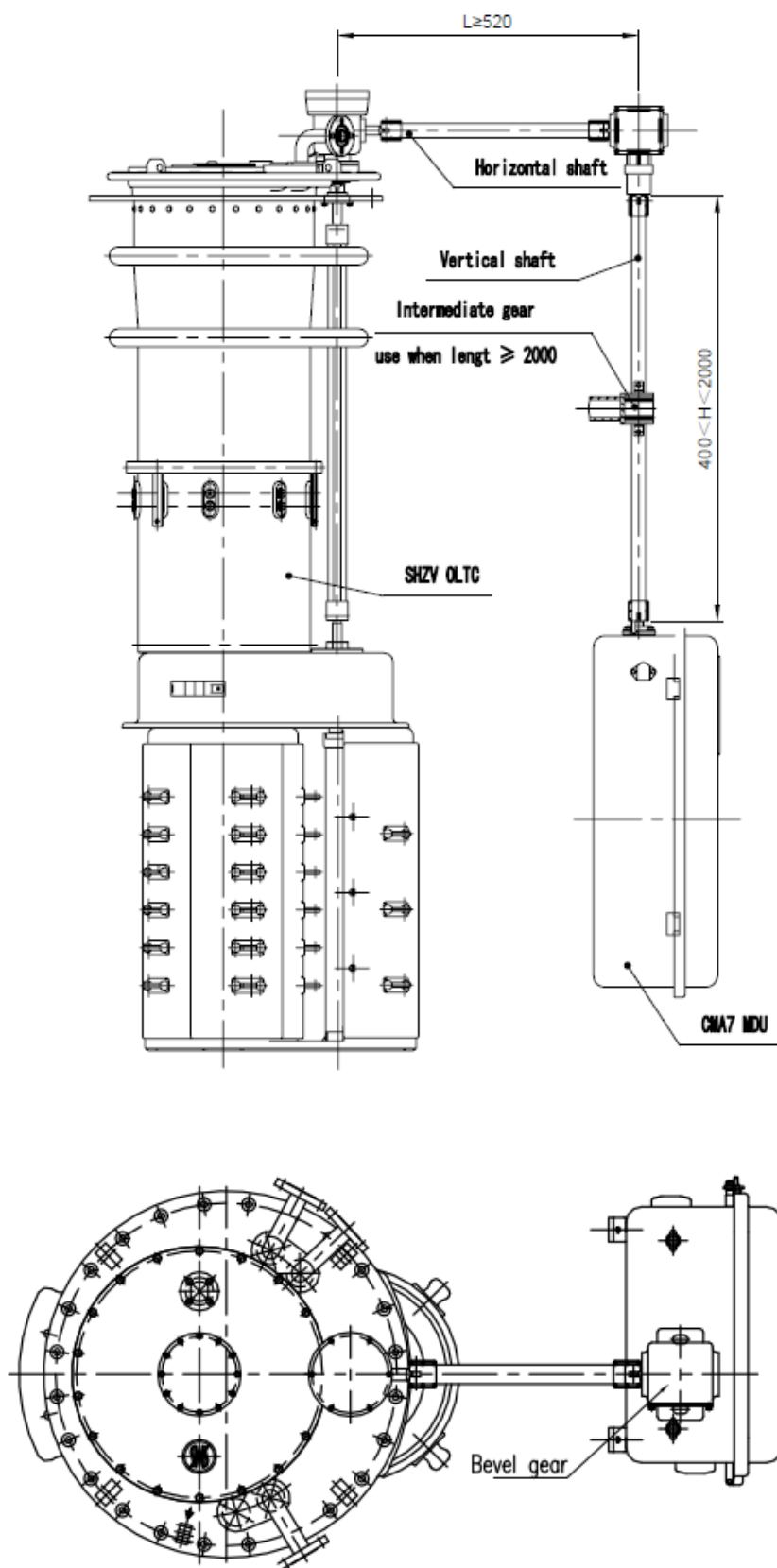
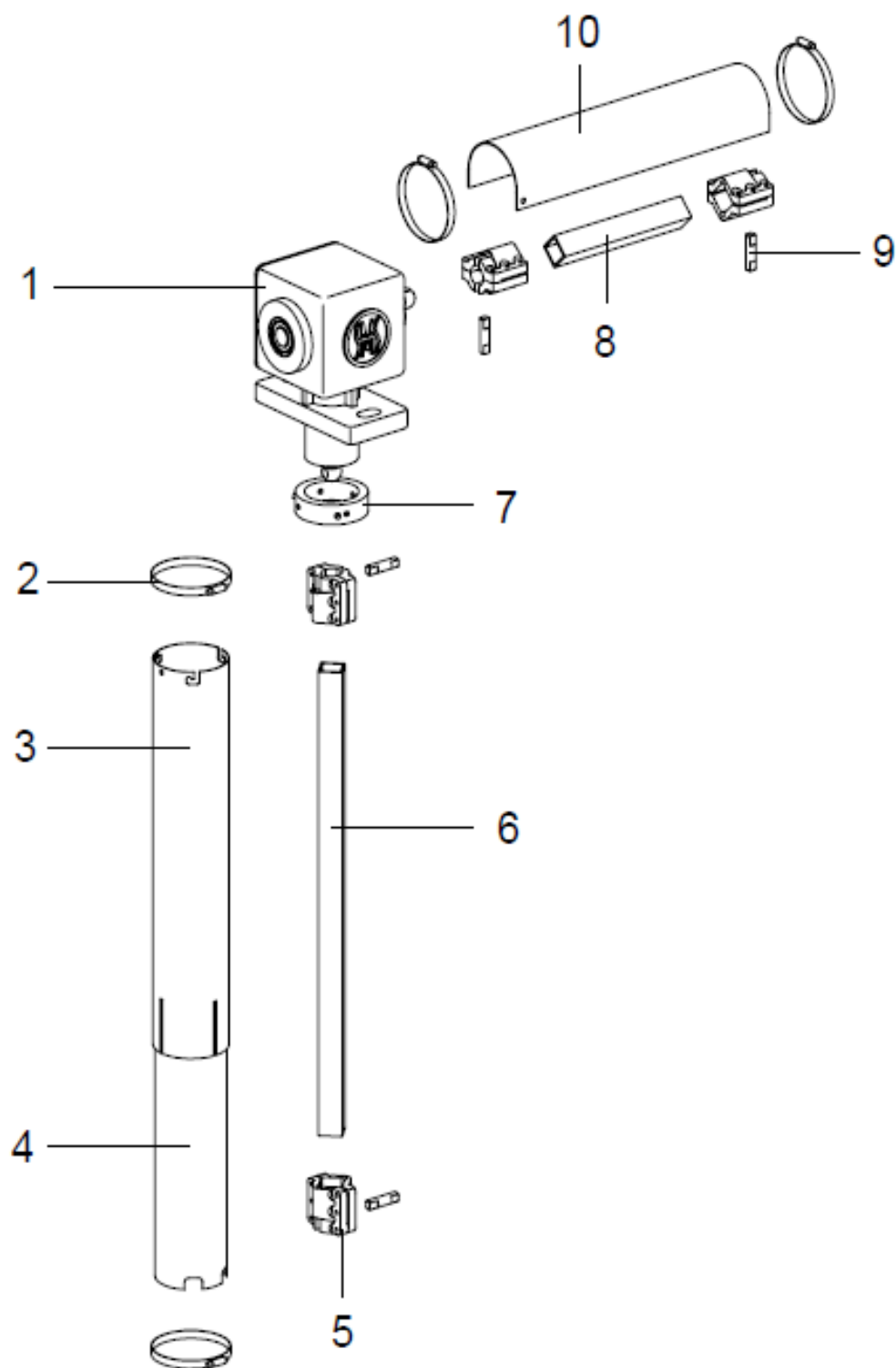


Figure 7.5.2

unit: mm

7.6 Installation of driving shaft protection

7.6.1 Installation procedure as figure 7.6.1



1.Bevel gear box	2.Hose clip
3.Telescopic protective tube 1	4.Telescopic protective tube 2
5.Coupling bracket	6.Vertical driving shaft
7.Adapting ring	8.Horizontal driving shaft
9.Coupling pin	10.Horizontal protective cover

Figure 7.6.1

7.6.2 Assembly of adapting ring



Prohibition

Install 3 stainless steel cylindrical pins on the adapting ring. Expose the pins around 5mm from outer ring, while exposing from the interior of the ring is prohibited.

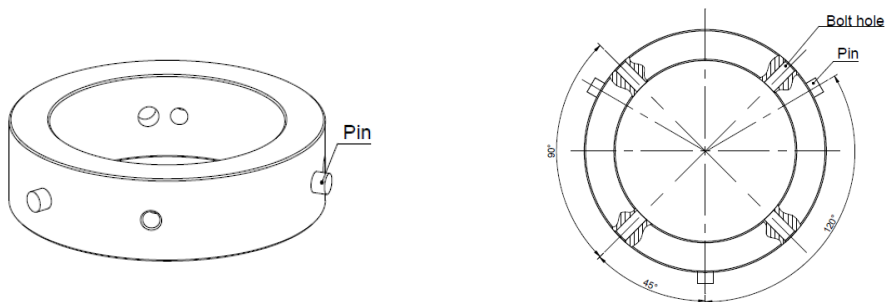


Figure 7.6.5 adapting ring

7.7 Installation of three single phase and I+II on-load tap-changer



In the process of interconnecting three single-phase and I+II on-load tap-changer, below points should be taken into consideration:

For three single-phase and the I+II arrangement, on-load tap-changers are located on the same transformer. since it will cause operation of diverter switch during the adjustment of top gear unit, so after adjustment, must check if each diverter switch is on same position and they have simultaneous action when operated by hand crank.

7.8 Calibration of on-load tap-changer driving unit

Manual operation for one cycle (full tap range) before electrical operation is required after connection between motor-drive unit and on-load tap-changer.

Procedure of connection verification, please refer to CMA7 motor-drive unit operation instruction.

8. On-load tap-changer routine test and test preparation

8.1 Test preparation

8.1.1 Fully bleeding on-load tap-changer

Before first commissioning, on-load tap-changer head and suction pipe on pipe connection must be bled.

8.1.1.1 Bleeding on-load tap-changer head

- 1) Remove the M30 screw cap on the bleeding valve E1 (use a 36mm wrench) on on-load tap-changer head cover.
- 2) Use screwdriver to lift valve tappet on bleeding valve E1 and bleed on-load tap-changer head (Figure.8.1.1) .
- 3) Seal the screw cap of the valve E1 (torque 9~12 N.m.)

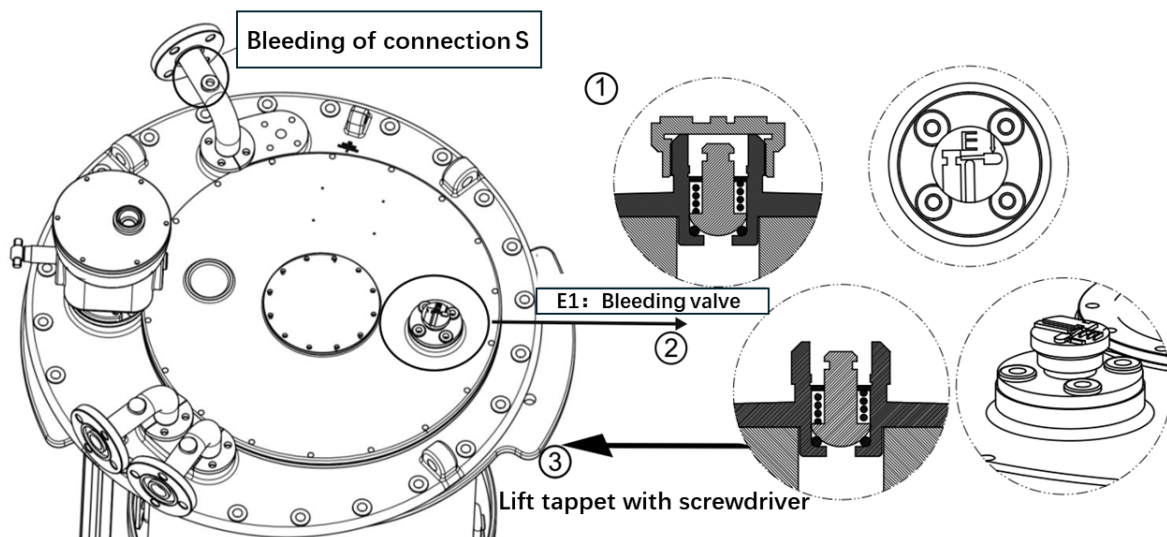
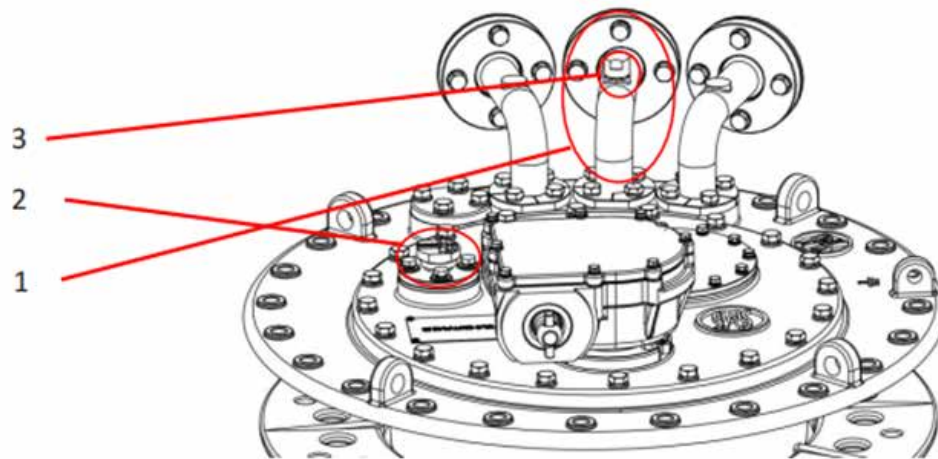


Figure 8.1.1

8.1.2 Bleeding suction pipe on the pipe connection S

- 1) Remove the M16 screw cap from the pipe connection S (use a No. 24 wrench (Figure.8.1.2).
- 2) Open vent screw M6 with slotted head and bleed pipe.
- 3) Close the vent screw (torque 2 N·m).
- 4) Seal vent screw with screw cap. (torque 8~10 N·m).



1. connection S 2. vent screw 3. M30 screw cap

Figure 8.1.2

8.1.2 Grounding

8.1.2.1 Connect the ground bolt of the tap changer head to the transformer cover (1 piece of M12 bolt and nut),

No. 19 wrench, torque 50~60 N·m, Figure.8.1.3.

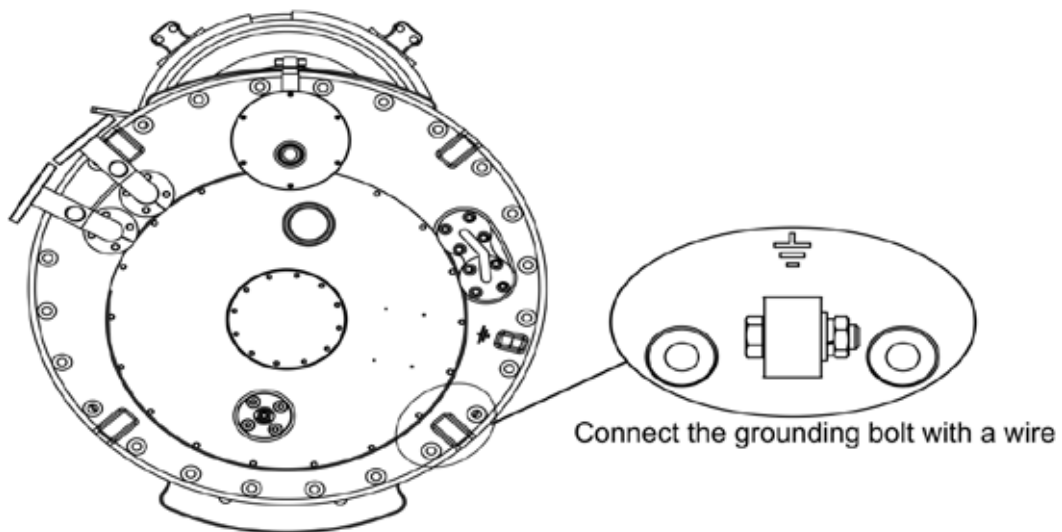


Figure 8.1.3

8.1.2.2 Connect the grounding bolt M12 on the motor-drive unit to the transformer tank (No. 19

wrench, torque 50~60 N·m).

8.2 Test in transformer factory

8.2.1 Operation test

Test tap-change operations must be carried out to check the mechanical functions of the on-load tap-changer and the motor-drive unit before the transformer energizing.



- 1) Before first actuating the on-load tap-changer after drying the transformer, the tap selector must be completely immersed in transformer oil and the diverter switch oil compartment must be completely filled with oil. Actuating the on-load tap-changer without oil will damage it.
- 2) During operation test, the on-load tap-changer must go through the entire operating range.
- 3) Make sure that the position indicators of the motor-drive unit and the on-load tap-changer (observation window on the on-load tap-changer head) read the same position in each operating position.



Warning

Warning of installation! (Danger of fatal or severe injury)

Serious damage on equipment and accident may cause!

If the connection between the on-load tap-changer and the motor-drive unit is improperly aligned, continuous operation will cause damage to the on-load tap-changer.

Ensure that the position indicators of the motor-drive unit and on-load tap-changer match.

8.2.2 The electrical test of the transformer

Flammable and explosive gases! (Danger of death or severe injury)

May cause Serious damage to equipment, death or injury to personnel, and lead to further accidents



Warning

Explosive gas may exist under the on-load tap-changer cover, in the connection pipes, oil conservator and the outlet of air dryer. Explosion may lead to the flying parts and the high temperature oil splash outside which will cause fatal and severe injury.

Deflagration warning! (Danger of death or severe injury)

May cause Serious damage to equipment, death or injury to personnel, and lead to further accidents



Warning

Please confirm: there is no open flame, heat source or electric spark (such as electric spark produced by static electricity) within its direct contact range, and there is no possibility of generating them. It must be ensured that the on-load tap-changer test must be carried out by professionals in accordance with the test procedures.

9. Transformer transportation and commissioning on site

9.1 Transport without motor-drive unit

9.1.1 Place the motor-drive unit in the setting position and then uncouple.

9.1.2 Remove the motor-drive unit.

9.1.3 Do not operate the motor-drive unit while the on-load tap-changer is not coupled. To refit, follow Huaming Operation instruction CMA7.

9.2 Oil fill and transport without oil conservator

If the transformer is filled with oil but stored and transported without oil conservator, a bypass connection must be installed between the interior of the diverter switch oil compartment and the transformer tank for pressure compensation. The bypass connection is installed between the pipe connection E2 and R.



NOTE!

For the storage period of 3 to 4 weeks without oil conservator, the oil level of the tap changer should be reduced by approx 5 liters.

9.3 Transport without oil

If the transformer is transported or stored without oil, the diverter switch oil compartment must be completely emptied. The interior of the diverter switch oil compartment should be protected in the same way as the transformer, such as by filling with nitrogen.



NOTE!

For long-term storage, the heater of the motor-drive unit should be regularly energized to prevent moisture.

9.4 Transformer commissioning on site



Warning

Deflagration warning! (Danger of death or severe injury)

Can cause serious damage to equipment, death or injury to personnel, and lead to further accidents

Make sure there is no open flame, heat source or electric spark (such as electric spark produced by static electricity) within 50m range, and there is no possibility of generating it.

**NOTE!**

Operation without oil will lead to damage of on-load tap-changer.

Before transformer commissioning, make sure the tap selector is completely immersed in the transformer oil and diverter switch oil compartment is completely filled with oil.

9.4.1 Operation on site:

9.4.1.1 When commissioning the transformer for the first time, make sure the transformer oil with Min Ud 40kV/2.5mm dielectric strength and Max.30 ppm moisture content.

9.4.1.2. Bleeding for on-load tap-changer please follow chapter 8.1.

9.4.1.3. Operation test please follow chapter 8.2.

9.4.1.4. Please make sure that the signaling contact for not reaching the minimum oil level in the the on-load tap-changer oil conservator is connected to the tripping circuit of the circuit breaker.

9.4.1.5. Check whether the protective relay function is normal according to the Huaming protective relay QJ-25 series instruction.

NOTE!

Once the transformer has been switched on, ensure that inrush current impulse has fully subsided before undertaking an on-load tap changing operation. The inrush current impulse are usually a multiple of the transformer rated current and can overload the on-load tap-changer during diverter switch operation.

Observe the safety instructions described in all functional checks and tests during the commissioning.

After the transformer has been switched on and the inrush current impulse has completely subsided, the transformer can be switched under no-load or load conditions.

10. Monitoring during on-load tap-change operation

10.1 Monitoring



CAUTION!

Visual inspection of on-load tap-changer head cover, protective relay and motor-drive unit should be carried out regularly.

Regulation type	Dielectric strength	Moisture content
At neutral point of winding	$\geq 30 \text{ KV}/2.5 \text{ mm}$	$< 40 \text{ ppm}$
At any points other than neutral point	$\geq 40 \text{ KV}/2.5 \text{ mm}$	$< 30 \text{ ppm}$

10.1.1 When the transformer is overloaded, on-load tap-changer cannot be operated frequently.

10.1.2 The trip contact of protective relay is set to operate when the oil speed is $1.0 \text{ m/s} \pm 10\%$. This contact should be connected to the tripping circuit of the circuit breaker; When the fault occurs inside the on-load tap-changer, a large amount of gas will be generated, resulting in accelerated oil flow rate, so that the baffle of protective relay is activated, trip contact closes, to cut off the transformer power supply and avoid the expansion of the accident. Once the protective relay is activated, it is strictly prohibited to switch on circuit breaker before lifting out diverter switch insert for check.

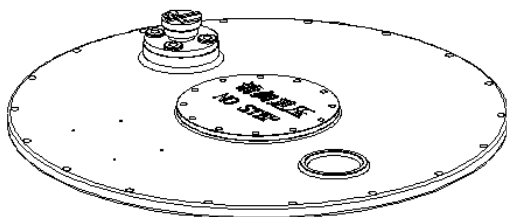
10.1.3 Rupture disk is installed on the on-load tap-changer head cover, which will not work during the normal operation of the tap-change. The rupture disk will break only if failure happened inside diverter switch oil compartment and the pressure in the oil compartment exceeds $0.3 \pm 20\% \text{ MPa}$ to avoid escalating the accident. During on-load tap-changer installation and maintenance, be careful not to step on or hit the rupture disk with heavy objects.

CAUTION!

No step on rupture disk!



CAUTION!



11. Notes for on-load tap-changer Maintenance

Electric shock warning! (Danger for Death or severe injury)

An energized transformer may cause death or severe injury.



Danger

- Turn off the high voltage and low voltage terminals of the transformer.
- Lock the transformer to prevent accidental restart.
- Ensure that all components are powered off.
- Ensure that all transformer terminals (earthing wires, earth disconnect switches) are earthed and short-circuited.
- Cover or isolate neighboring live parts.

Electric shock warning! (Danger for Death or severe injury)

Operation of energized on-load tap-changers may cause personnel death or severe injury.



Danger

- Disconnect all auxiliary circuits, such as tap changer monitoring devices, pressure relief valves and pressure monitors.
- Ensure that all components are powered off.

Warning! (Danger for Death or severe injury)

Risk of explosion!



Danger

- Explosive gas in the diverter switch oil compartment, transformer tank, connection pipes, oil conservator, and the outlet of the dehydrating breather can ignite or explode, resulting in severe injury or death.
- Ensure that there are no open flames, hot surfaces or electrical sparks (e.g. caused by static build-up) in the environment surrounding the transformer.
- Do not operate any electrical devices (for example, impact wrenches have the risk of spark).
- Only use conductive and grounded hoses, pipes and pump-operated equipment which are suitable for flammable liquids.

Check the motor-drive unit!

Condensation in the motor-drive unit can cause damage to the motor of the motor-drive unit.



- Always keep the cabinet of motor-drive unit sealed.
- If the motor-drive unit is out of work for more than 2 weeks, connect and operate the anti-condensation heater in the motor-drive unit. If the condition is not allowed (for example, the heater cannot be energized during transportation), a sufficient amount of desiccant must be placed inside the cabinet.

12. Inspection and maintenance

12.1 Regular inspection

Please follow the inspection instruction below:

Interval	Instruction	Details
Once per year	Check the door sealing, cable sleeves, motor-drive unit cabinet and control panel's ventilation device	Visually check. If the sealing parts are aging and broken, and there is oil leakage, replacement is required immediately. The ventilation of the motor-drive unit and control panel should be tested. If aging damage occurs, replacement is required immediately.
Once per year	Check the sealing of on-load tap-changer top cover, protective relay and all pipe connections.	Visually check. If the sealing parts are aging and broken, and there is oil leakage, replacement is required immediately.
Once per year	Check if the heater inside the motor-drive unit cabinet works properly.	Visually check. If the electronic components are getting black or yellow, and the pins are broken, please contact Huaming service. Then switch on temperature controller and check its functionality use a thermometer.
Once per year	Check the protective relay .	Visually check. If the electronic components are getting black or yellow and the pins are broken, please contact Huaming service.
Once per year	Check if the silica gel in the breather of on-load tap-changer oil conservator is working properly	Put the humidity indicator card in the oil conservator for at least half an hour, if the humidity is greater than 10%, replace the silica gel, or replace it every other year.
Once per year	Check the oil sample.	Discharge the transformer oil inside pipe for around 5L, then take the oil sample from diverter switch. If the parameter of the oil sample does not meet with the data in Table 12.1.2, it is recommended to filter or replace the transformer oil.

Table 12.1.1 Inspection instruction

Item		Dielectric strength	Water content
Limit values for insulating oil in accordance with IEC 60296			
When commissioning the transformer for the first time		> 60kV/2.5mm	< 15ppm
During operation	Class I tap-changer	> 30kV/2.5mm	< 40ppm
	Class II tap-changer	> 40kV/2.5mm	< 30ppm
After maintenance		> 45kV/2.5mm	< 15ppm
Limit values for natural esters in accordance with IEC 62770			
When commissioning the transformer for the first time		> 60kV/2.5mm	≤ 100ppm
During operation	Class I tap-changer	> 30kV/2.5mm	≤ 200ppm
	Class II tap-changer	> 40kV/2.5mm	≤ 200ppm
After maintenance		> 45kV/2.5mm	≤ 100ppm

Table 12.1.2 Oil sample limit values

12.2 Maintenance :

Interval	Instruction
After 300,000 times on-load tap-changer operation (based on the number shown in motor drive-unit's counter)	on-load tap-changer inspection and maintenance. Please contact Huaming's service department.
After 600,000 times on-load tap-changer operation (based on the number shown in motor drive-unit's counter)	Inspection for vacuum interrupters. Please contact Huaming's service department.
After 800,000 times on-load tap-changer operation (based on the number shown in motor-drive unit's counter)	Repalce the diverter switch insert. Please contact Huaming's service department.
After 1,500,000 times on-load tap-changer operation (based on the number shown in motor-drive unit's counter)	Tap selector maintenance. Please contact Huaming's service department.

Table12.1.3 Maintenance interval



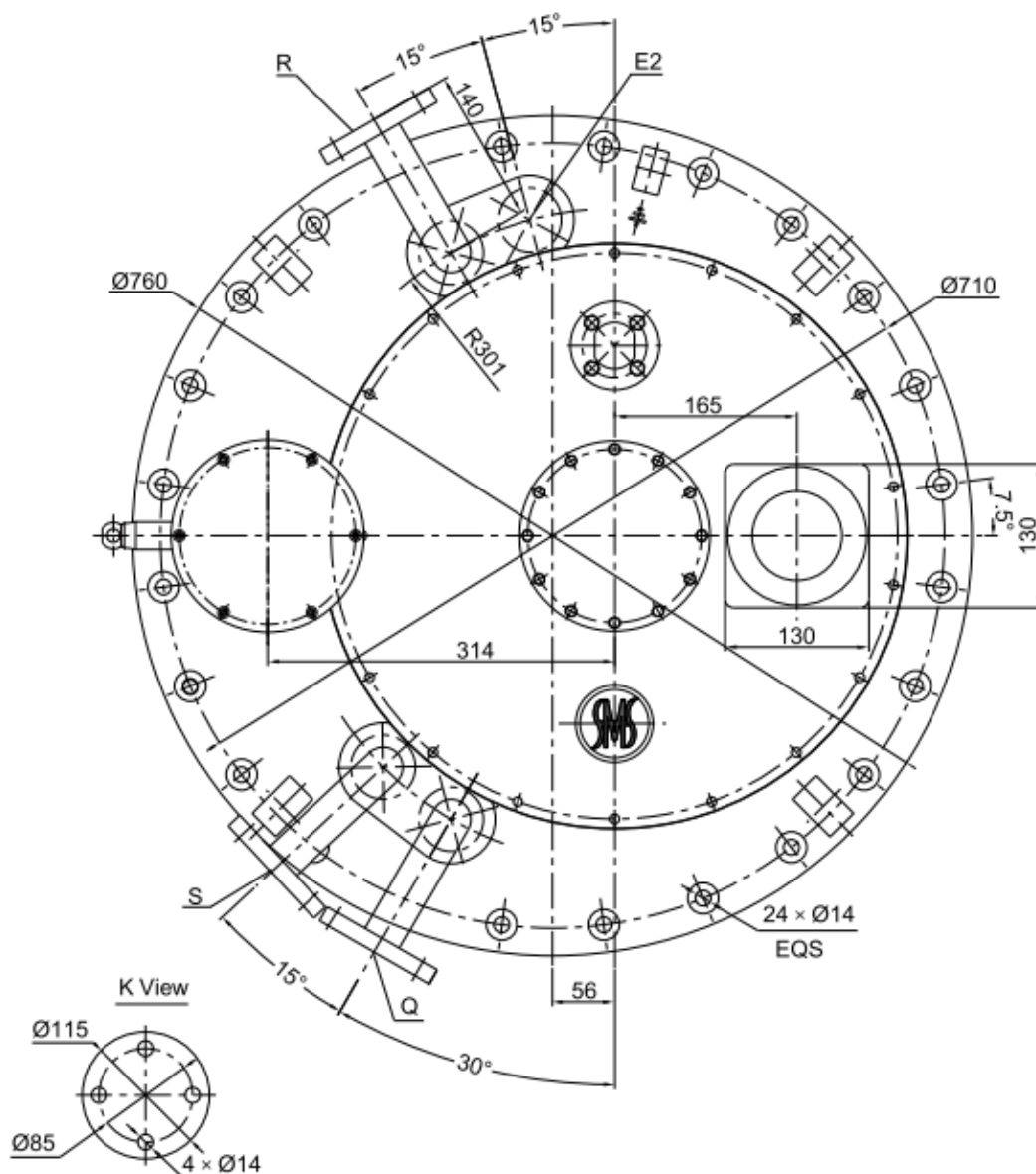
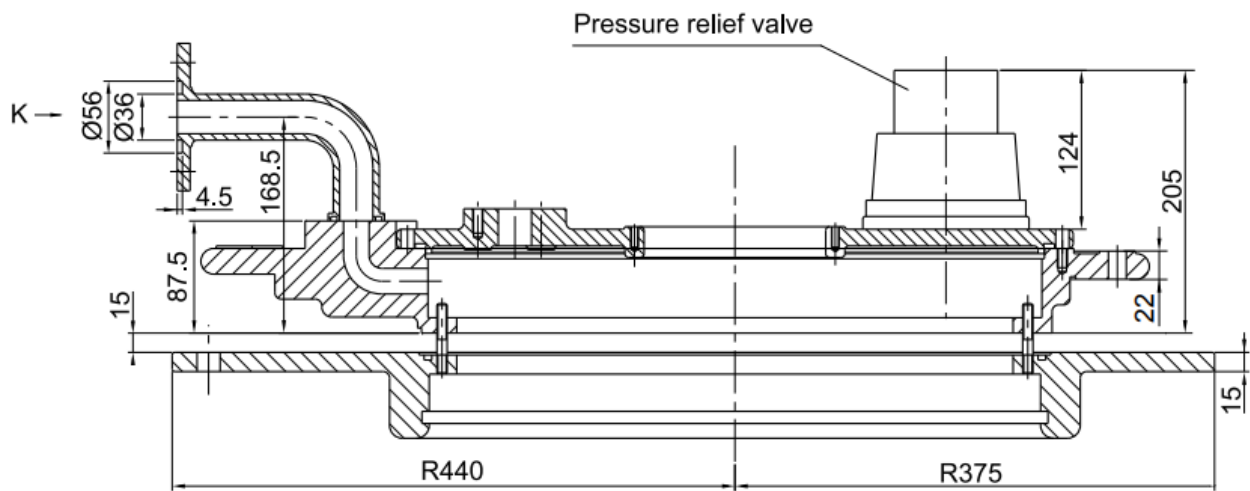
Damage to on-load tap-changer and transformer for non-observance
of maintenance interval and improper maintenance.

Please write to export@huaming.com for any support you may need.

13. Appendix

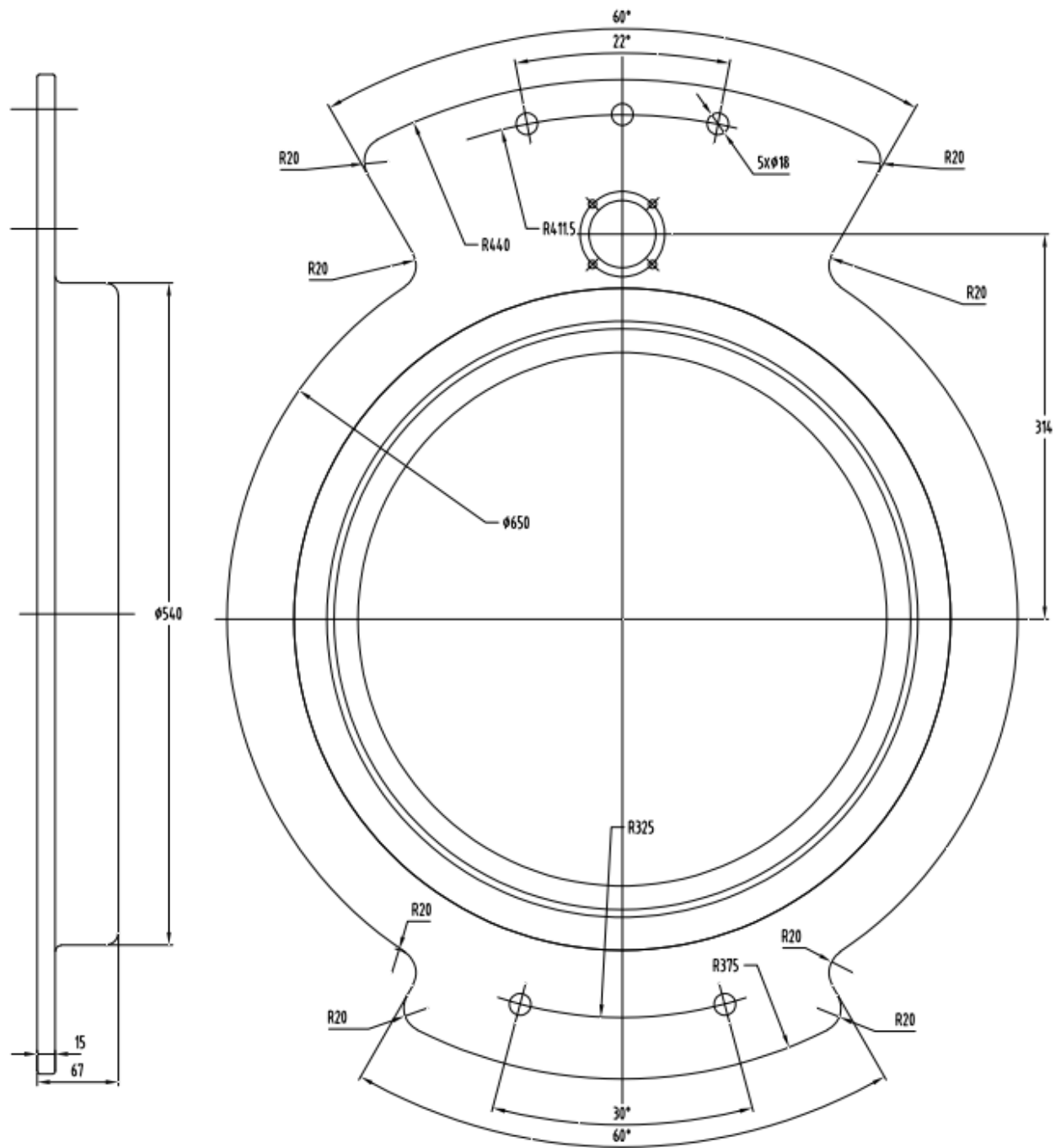
Appendix 1. Overall dimension of SHZV head flange	53
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Appendix 2. Overall dimension of SHZV head flange with pressure relief valve



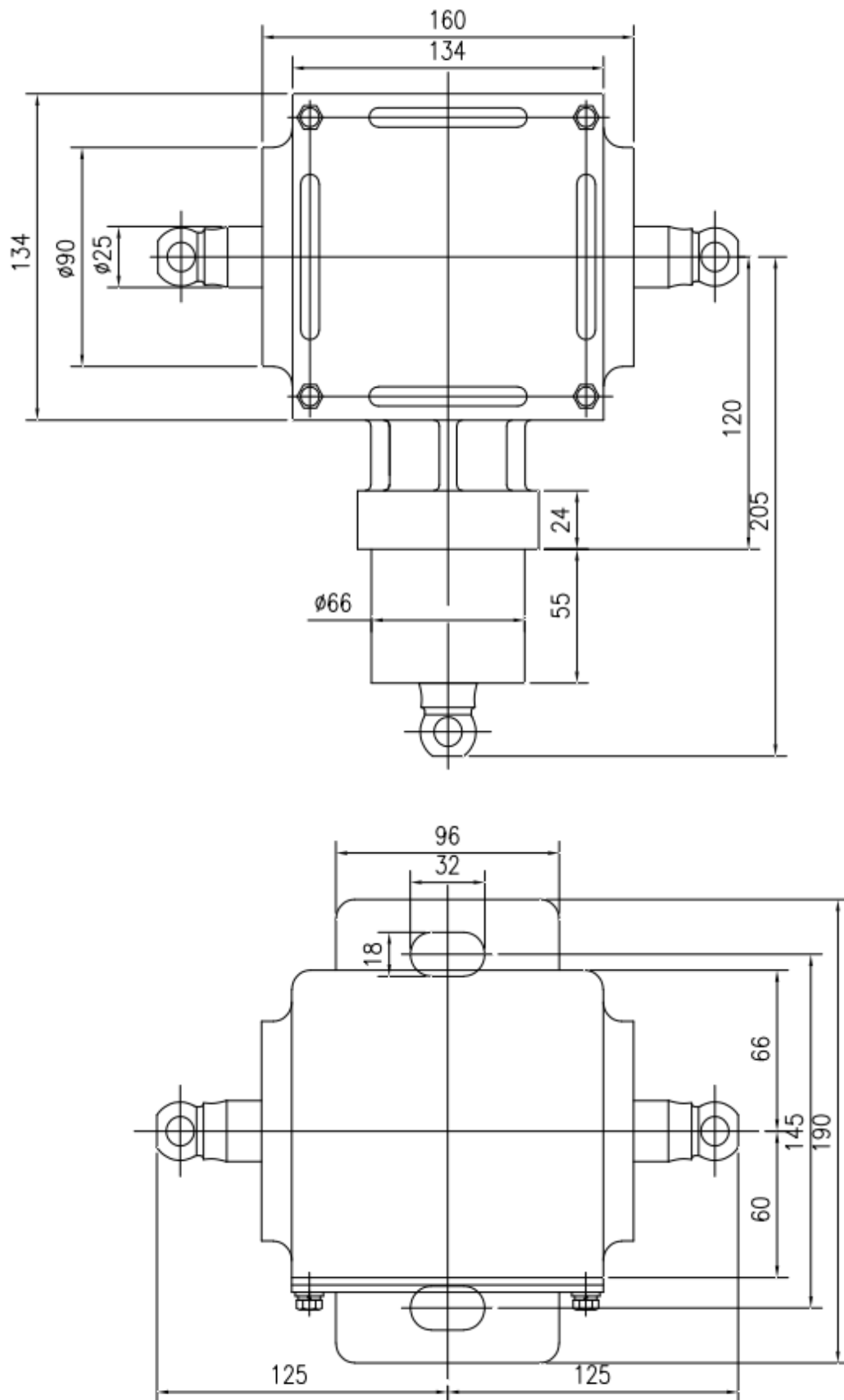
Unit:mm

Appendix 3. Overall dimensions of supporting flange for SHZV on-load tap-changer



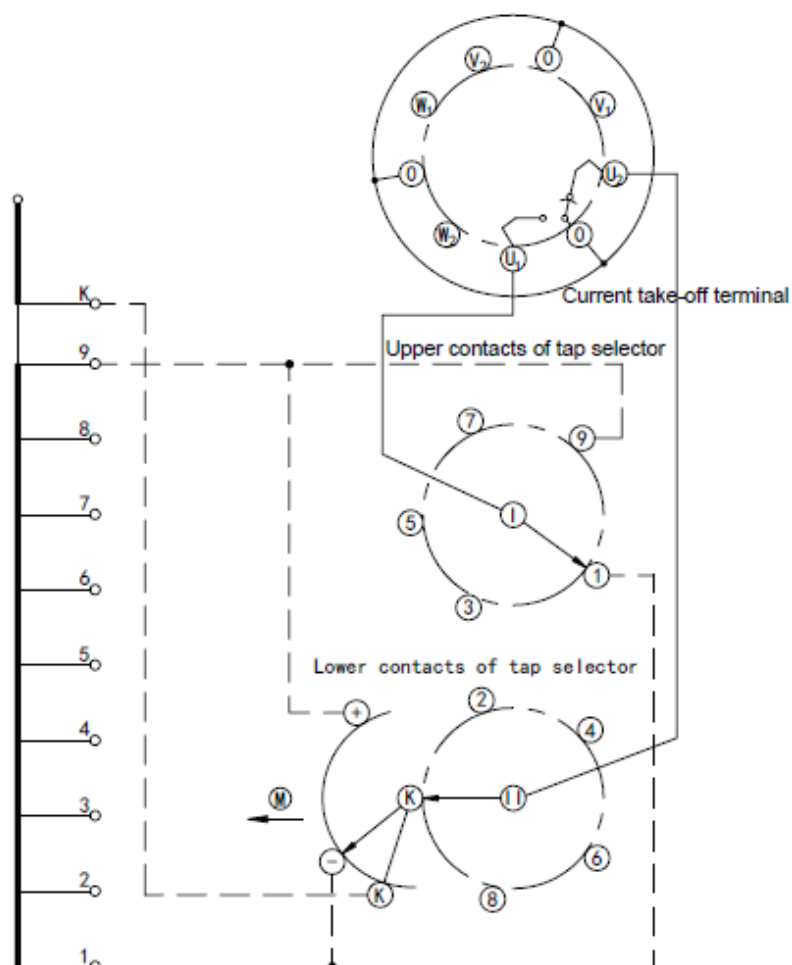
unit: mm

Appendix 4. Overall dimension of bevel gearbox

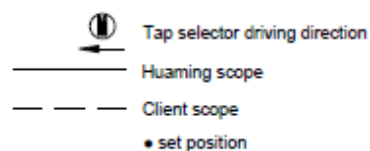


unit: mm

Appendix 5. SHZV (10193W) operating position table and connection diagram



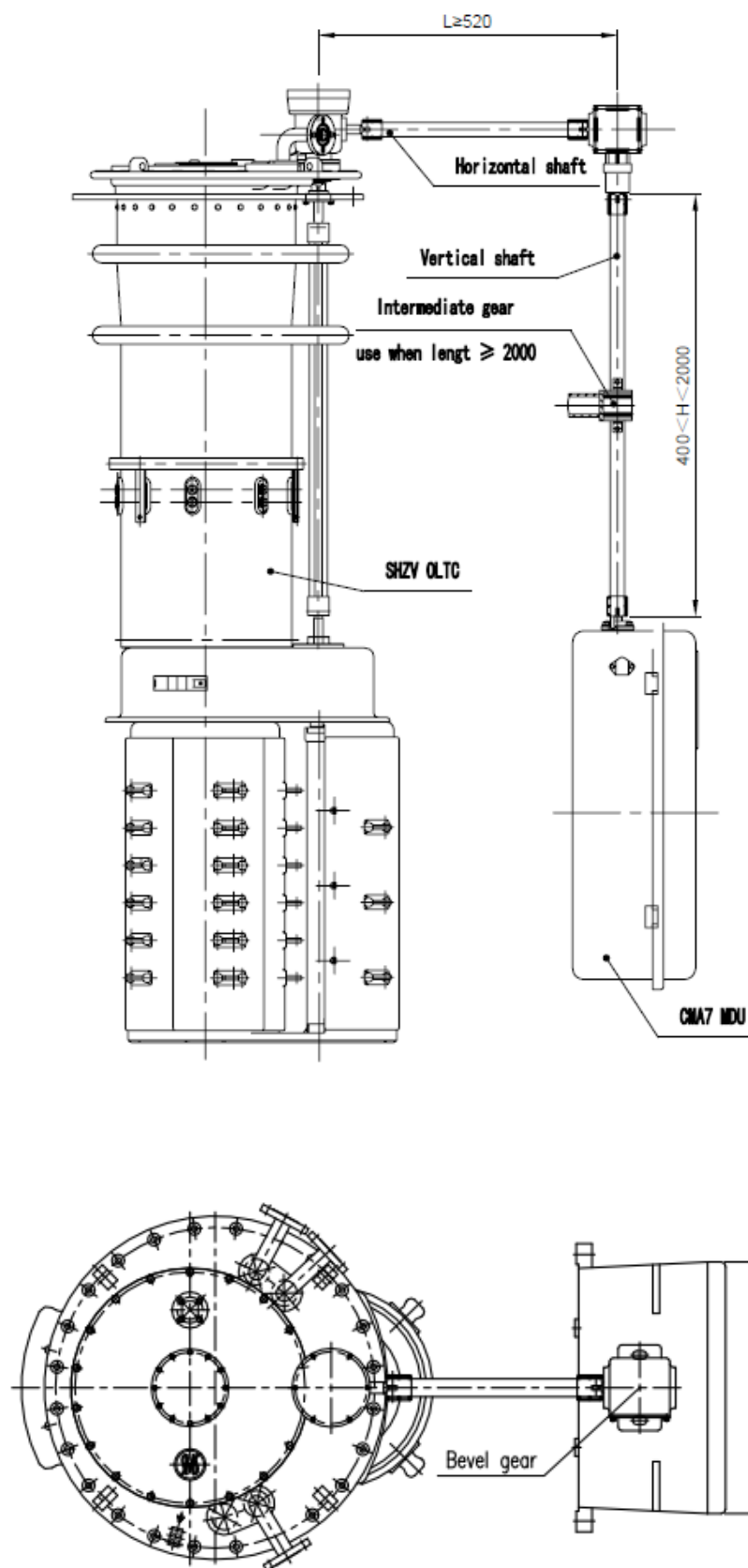
Operation position number	19
Different voltage number	17
Set position●	9b



Change-over selector																			
Tap selector contact position	1	2	3	4	5	6	7	8	9	K	1	2	3	4	5	6	7	8	9
Display position	1	2	3	4	5	6	7	8	9a	9b	9c	10	11	12	13	14	15	16	17

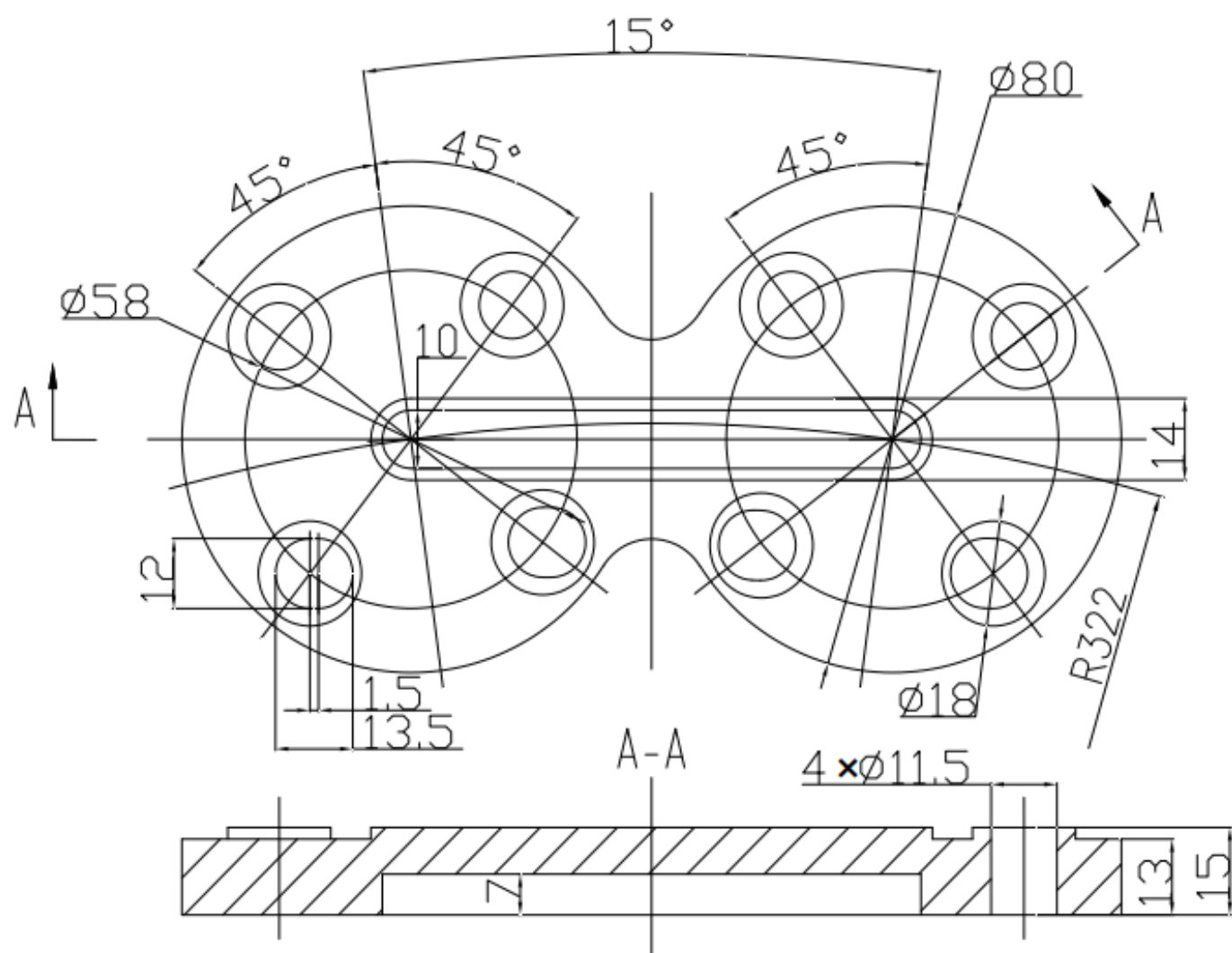
unit: mm

Appendix 6. Schematic drawing for connection of SHZV and CMA-7



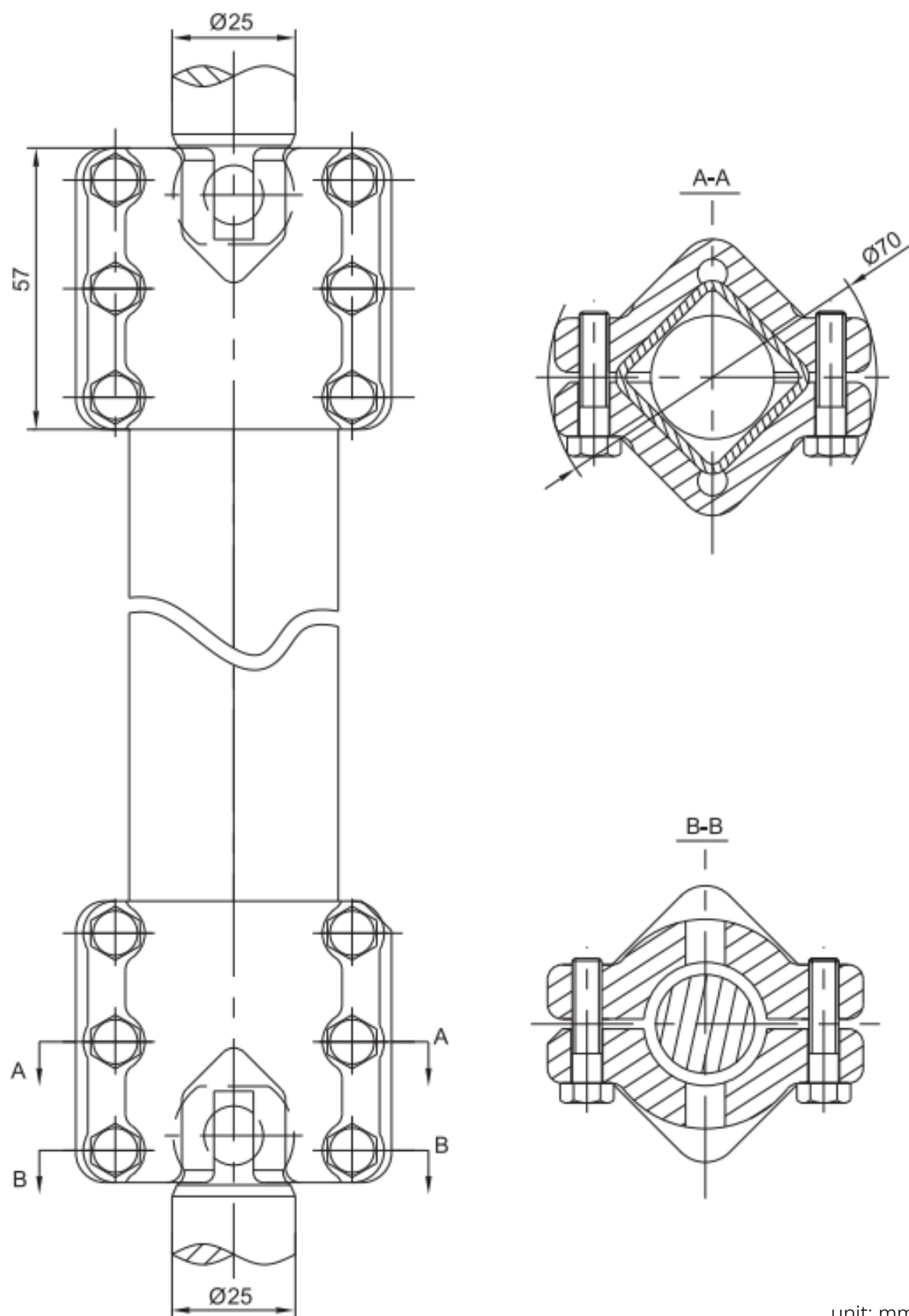
unit: mm

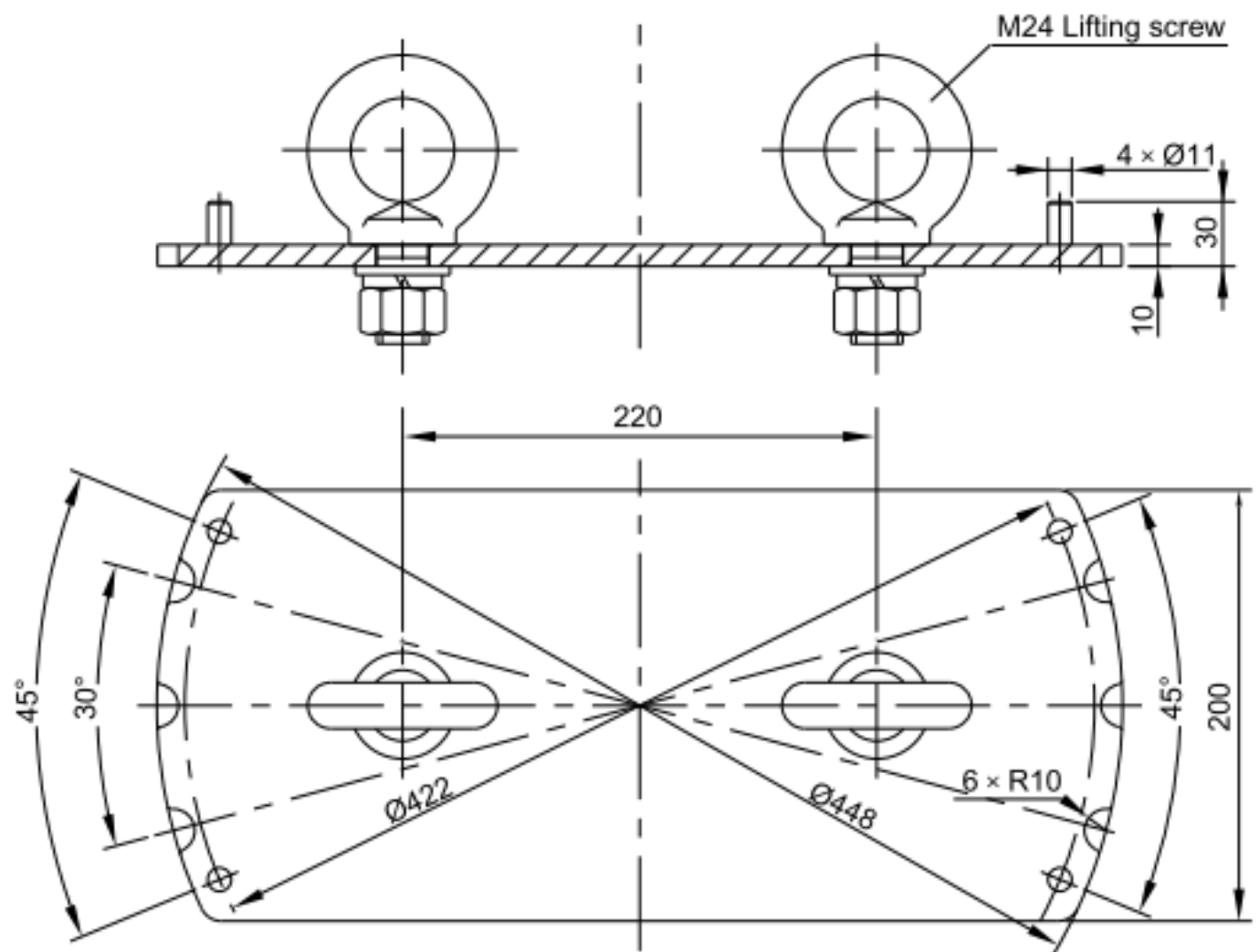
Appendix 7. Overall dimension of by-pass connection



unit: mm

Appendix 8. Schematic drawing for connection of horizontal shaft and vertical shaft



Appendix 9. Overall dimension of lifting plate

unit: mm



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